Learning Verified Monitors for Hidden Markov Models

Luko van der Maas and Sebastian Junges

Radboud University, Nijmegen, the Netherlands

Abstract. Runtime monitors for hidden Markov models obtain a sequence of observations and must classify whether the system is currently deemed safe. A system is deemed safe, if the probability of being in a dangerous state conditioned on these observations exceeds some acceptable threshold. The key contribution of this paper is the first correct-by-construction synthesis method for such monitors, represented as finite state machines. First, we establish the coNP-hardness of checking whether a finite state machine is indeed a correct monitor, i.e., a monitor without misclassifications. Second, we provide a reduction that reformulates the search for misclassifications into a standard probabilistic system synthesis problem. Third, we integrate the verification routine into an active automata learning routine to synthesise correct monitors. Fourth, we provide a prototypical implementation that shows the feasibility and limitations of the approach on a series of benchmarks.

1 Introduction

Runtime assurance is an essential ingredient in the deployment of safe autonomous systems. Runtime monitors provide assurance by flagging potentially dangerous system behavior by observing a system execution. More precisely, a monitor receives a stream of observations about the system and outputs a verdict, e.g., it raises an alarm that the system has left some safety envelope. A monitor is correct if it correctly raises such alarms based on a formal specification. Various challenges in creating correct runtime monitors for (semi-)autonomous systems have been identified [**empty citation**], such as: (1) the state of the system is only partially observable, i.e., the stream of observations comes from sensor readings and does not uniquely identify the state of a system, (2) the behavior of the system and/or the sensors may be subject to probabilistic uncertainty, (3) the monitor itself is subject to resource constraints (time, memory, etc), and (4) the monitor is itself safety-critical and should therefore be subject to extensive validation. Challenges (1,2) can be (partially) addressed by modelling the system as a hidden Markov model (HMM), Challenges (3,4) can be (partially) addressed by representing a monitor as, e.g., a (small) finite state machine. Concretely, this paper focuses on the following question: *Is a given finite state machine a correct monitor for a given HMM?* The paper studies the complexity of this problem, provides a practical verification procedure, and embeds this verification procedure into a learning framework.

Linus: What does partially mean here? Perhaps a footnote that says you discuss this somewhere?

¡ *What are HMMs?* In this paper, we assume that the system including its sensors is adequately described as a discrete Hidden Markov Model (HMM). Markov chains (MCs) describe system behavior subject to probabilistic uncertainty. Paths through an MC are sequences of states that describe system executions. HMMs extend MCs by labelling their states with colors (aka *observations*). Intuitively, the observations can be used to model the information that the monitor receives. In HMMs, every path can be lifted to a sequence of observations, which we call a *trace*. The trace associated to a system execution describes the information received by the monitor.

Linus: Maybe move after What are correct monitors. I feel like "what are correct monitors" is a question that immediately comes up, while risk of a trace is not.

Linus: How does this relate to the problem statement?

Risk of a trace. In runtime monitoring, an essential task is to estimate whether the current system is dangerous. In the simplest setting, a specification describes whether a set of states is dangerous. The risk of a trace is then the probability of ending in a dangerous state, conditioned on the fact that the system execution matches the trace. For a given trace, we may compute this risk, e.g., either via a reduction to model checking [**empty citation**] or by a (forward) filtering that tracks a distribution over the current states [**empty citation**].

What are correct monitors? Often, we are not interested in this precise risk, but rather want to partition the set of horizon-bounded traces into safe and unsafe traces: Unsafe traces are traces whose risk does exceed some fixed *acceptable threshold*, all other traces are safe. Indeed, a *correct* monitor obtains a trace and raises an alarm iff a trace is not safe. The key verification problem in this paper asks whether a monitor, represented as a deterministic finite automaton, accepts (i.e., raises an alarm) exactly the unsafe traces. We note that in this paper, we only consider traces that are bounded by some fixed horizon.

Illustrative toy example. SJ: TODO make more concrete, add Fig. As an running example, we consider an oversimplified car driving scenario, loosely inspired by runtime monitors obtained from high-fidelity simulations [**empty citation**]. A car can be in three states: It can be on a dry road, on an icy patch, or it can drifted off the road. The HMM in Figure 3 describes the situation, where a car alternates between dry and icy road segments, and where being on a icy (dry) segment positively affects the probability that the next segment is icy (dry). When on an icy road, there is a higher probability to drift off the road. Our sensor can detect dry roads, but cannot distinguish between icy roads and off-road conditions. We can define the trace risk as the probability to be off-road, conditioned on a a trace, such as $dry \cdot icy \cdot icy$. In Fig. 3b, monitor $\mathcal{A}$, we depict a finite automaton that describes the set of traces that end with two consecutive icy patches. The key verification question in this paper is now: Is this a correct monitor for threshold 0.9? Put differently, does it raise an alarm iff the trace risk is above 0.9.

SJ: I parked this figure here from the proposal

SJ: Do we also want a small figure of a system feeding observations into a monitor?

Novel complexity results. Deciding whether a monitor is correct is coNP-complete (Theorems 4). In particular, we study the dual problem that asks whether a

monitor misclassifies at least one trace. We consider a simpler variant of this problem that asks whether there is a bounded trace in the HMM that is unsafe. [1] The problem is clearly in NP (assuming that the bound is unary encoded). We show that this simplified problem is also strongly NP-hard (Lemma 5) and the corresponding optimization problem, that asks to approximate the risk of the trace with the heighest risk, is APX-hard (Lemma 6). Intuitively, APX-hardness means that the existence of an efficient approximation algorithm would have major complexity-theoretic implications.

Our verification approach. While deciding whether a monitor is correct is thus in general not tractable, we suggest to utilise recent advances in the synthesis of probabilistic systems. We concentrate on proving the absence of missed alarms, i.e., we concentrate on showing that the monitor correctly identifies every unsafe trace. A similar reduction works for showing the absence of spurious alarm. First, for a single trace, computing the risk can be reduced to computing a reachability probability in a MC that is intuitively a kind of product between a DFA that accepts exactly the trace and the HMM [**empty citation**]. Thus, it is natural to reduce our problem to the question: Is there a DFA (accepting exactly one trace, accepted by our monitor) such that the probability in the product MC exceeds a threshold. If the answer is no, then the monitor has no-missed-alarms. This decision problem can be solved using (exact) probabilistic model synthesis methods [**empty citation**] as implemented in the tool Paynt [**empty citation**][2]. We discuss our verification approach in Seciton 3.

Linus: The term spurious alarm only appreas here, maybe describe it instead?

Linus: emph?

Our learning approach. Ultimately, we do not only want to verify the correctness of a monitor, but we want to construct such a monitor. Above, we already mentioned that the monitors are formal languages. When considering bounded trace, these monitors are regular, and can thus be captured using DFAs. Inspired by a recent short paper [**empty citation**], we investigate learning these monitors: We choose to do this using active automata learning [4, 12], which closely relates

Linus: This was not mentioned yet

[1] The results also apply to checking for safe traces.

[2] Alternatively, in Section **??**, we show that we can construct a product of an acyclic DFA accepting sets of traces and the HMM. However, this yields partially observable MDP, where we must search for memoryless policies, similar to a construction in [5]. This problem is equivalent to the synthesis problem and can therefore also be solved using Paynt [2].

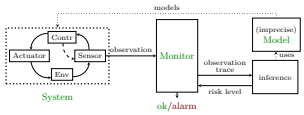


Fig. 1: White-box monitoring of systems

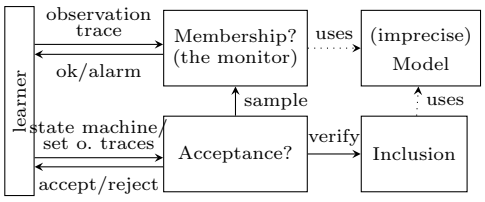
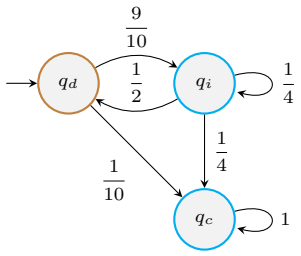
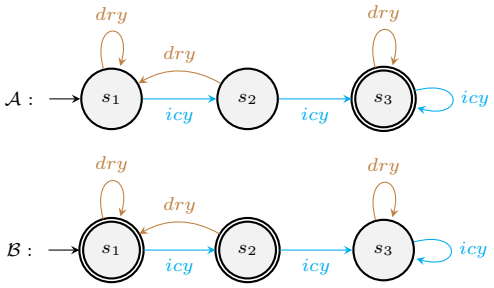


Fig. 2: Learning monitors.



(a) Example HMM, where the cyan states q_i and q_c have the observation *icy* and the brown state q_d has the observation *dry*.



(b) Two example monitors for the HMM in Fig. 3a, correct monitor $\mathcal{A}$ and faulty monitor $\mathcal{B}$. $\mathcal{A}$ raises an alarm as soon as two consecutive steps on an *icy* section have been observed. While $\mathcal{B}$ raises an alarm if less than two consecutive steps on *icy* road has been observed.

Fig. 3: The central example model as a HMM with both a correct monitor and a faulty monitor for a threshold of 0.8.

to other popular oracle-guided inductive synthesis loops [**empty citation**]. To use active automata learning (AAL), we must provide a *membership oracle* that decides for a single trace whether the risk of the trace exceeds a threshold and an *equivalence oracle* that decides whether a given hypothesis monitor is indeed correct. The membership oracle can be implemented by computing the risk for a single trace as above and comparing it to a threshold, while the equivalence oracle can be implemented by checking for the absence of missed and false alarms. Practically, some minor modifications are necessary to embed our approach in AAL framework: We have to handle the finite horizon, resolve some traces that are not relevant for the monitor, and we combine conformance and equivalence queries to boost performance.

Putting it together. In Fig. **??**, we outline our proposed framework to learn a monitor. SJ: TODO explain framework

Empirical evaluation. SJ: TODO summarise results of the empirical eval here.

Contributions. In summary, this paper provides a first framework to learn runtime monitors — encoded as finite automata — that are verified to be correct with respect to a given HMM. The main contributions are as follows: (1) We study monitor verification and show complexity-theoretic hardness of the underlying problem, even in an approximative setting. (2) We reduce the monitor verification to probabilistic model synthesis problems that exhaustively search for a counterexample. (3) We learn monitors by using the verifier above to answer equivalence queries in conjunction with active automata learning. (4) We demonstrate the feasibility and limitations of both the verifier and the learner on a series of benchmarks. Our prototype can handle HMMs with thousands of states, up to a hundred observations, and monitors with over 100 states.

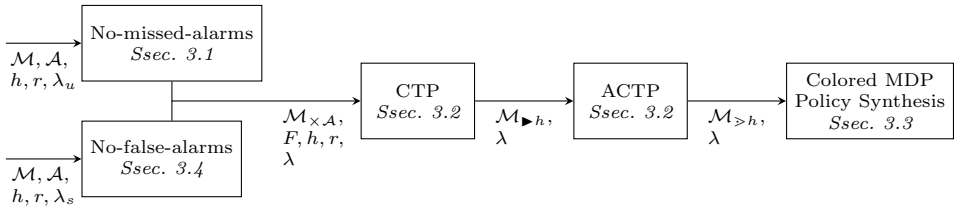


Fig. 4: Overview of proof of Thm. 1, containing the different intermediate problems, together with the inputs for each problem.

2 Problem Statement

In this section, we recap standard notions on stochastic models and finite automata and introduce the formal problem statement.

Notation. Let X be a finite set. A distribution μ over X is a mapping $\mu\colon X \to [0,1]$ such that $\sum_{x \in X} \mu(x) = 1$. The set of all distributions over X is written $\Delta(X)$.

2.1 Models

In this section, we briefly introduce MDPs, which are an integral part of our approach, and HMMs which are crucial to the problem statement. Details can be found in [6].

Definition 1 (MDP). *A* Markov decision process *(MDP) is a tuple* $\mathcal{M} = (S, \iota, Act, \mathbf{P})$ *with a countable nonempty set* S *of* states*, the* initial state $\iota \in S$*, the* partial transition function $\mathbf{P}\colon S \times Act \rightharpoonup \Delta(S)$.

We use $Act(s) = \{a \mid \mathbf{P}(s,a) \neq \bot\}$ as the set of enabled actions. We assume that there are no deadlocks, i.e., for every $s \in S$, $Act(s) \neq \emptyset$. A path π in an MDP is a sequence $s_0 a_0 s_1 \ldots \in (S \times Act)^* \times S$, such that $a_i \in Act(s_i)$ and $P(s_i, a_i)(s_{i+1}) > 0$ for every $i \geq 0$. The set of (finite) paths is denoted by $\Pi^{\mathcal{M}}_{fin}$, the set of paths of at most length h are $\Pi^{\mathcal{M}}_{h}$, and the set of paths of exactly length h are $\Pi^{\mathcal{M}}_{=h}$. If the MDP is clear from the context, the $\mathcal{M}$ is dropped.

A policy for an MDP is a function $\sigma\colon \Pi^{\mathcal{M}} \to Act$, which given a path through the MDP, gives a action. Given a policy, a MDP can be induced into a MC. Policy synthesis for an MDP for a property ϕ thus entails finding a policy for an MDP such that the induced MC entails ϕ.

A *Markov chain* (MC) is an MDP where $|Act(s)| = 1$ for every state $s \in S$. We simplify notation and write MCs also as tuple $(S, \iota, \mathbf{P})$ and write $\mathbf{P}(s)$ to refer to the unique distribution $\mathbf{P}(s,a)$. Paths in a MC can be simplified to be sequences of (only) states.

The probability measure $Pr^{\mathcal{M}}$ of an MC $\mathcal{M}$ is the unique probability measure following from the canonical σ-algebra associate with mc.

Luko: Condition probability on traces defined using Paths

Definition 2 (HMMs). *A (risk-labelled) HMM is a tuple* $(S, \iota, \mathbf{P}, Z, obs, r)$ *such that* $(S, \iota, \mathbf{P})$ *is a MC,* Z *is a finite set of observations, and* $obs\colon S \to Z$ *is the (deterministic) observation function*[3]*, and* $r\colon S \to \mathbb{R}_{\geq 0}$*,*

Notions such as paths are lifted from MCs. Furthermore, traces τ are sequences of observations, and the set of traces of a HMM, for which there is non-zero probability path, is the language $\mathcal{L}(\mathcal{M})$ of the HMM. The set $\mathrm{Paths}(\tau)$ contains all paths that correspond to a trace τ, $\mathrm{Paths}^{\mathcal{M}}(\tau) := \{\pi \in \Pi_{\mathcal{M}} \mid obs(\pi) = \tau\}$. The ith observation of a trace is denoted as $\tau^{(i)}$. The condition probability on a trace $\tau \in \mathcal{L}(\mathcal{M})$ is defined using Bayes' rule

Luko: Premise paper says characterized not defined, why?

$$Pr(\pi \mid \tau) := \frac{Pr(\tau \mid \pi) \cdot Pr(\pi)}{Pr(\tau)}$$

Where, $Pr(\tau \mid \pi)$ is 1 if $obs(\pi) = \tau$ and the probability of a trace τ is $\sum_{\pi \in \Pi_{\mathcal{M}}} Pr(\pi) \cdot Pr(\tau \mid \pi)$.

Luko: Fill in example

Example 1.

Definition 3. *A deterministic finite automaton (DFA) is a tuple* $\mathcal{A} = (Q, \Sigma, \delta, \iota, F)$*.* Q *is a finite set of states,* Σ *is an alphabet,* $\delta\colon Q \times \Sigma \rightharpoonup Q$ *is a (partial) transition function,* ι *is the initial state, and* $F \subseteq Q$ *is a set of accept (or: final) states.*

Let ε denote the empty word. We lift the transition relation to words, which is defined recursively: $\delta^*(q, \varepsilon) = \varepsilon$ if $a \in \Sigma$ and $\delta^*(q, w \cdot a) = \delta(\delta^*(q, w), a)$.

The *language* $\mathcal{L}(\mathcal{A})$ of a DFA $\mathcal{A}$ is the set of all words accepted by the $\mathcal{A}$, $\mathcal{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \delta^*(\iota, w) \in F\}$.

2.2 Formal Problem Statement

The monitor verification problem consists of two sub-problems: the no-false-alarms problem, and the no-missed-alarms problem. Both of these sub problems entail finding the trace with either the minimum or maximum risk. We will discuss the no-missed-alarms problem in the following sections. In Ssec. 3.4 we will adapt to the no-false-alarms problem.

The no-missed-alarms problem entails ensuring that all traces from the HMM, which are not in the monitor, have at most a risk of λ.

Definition 4 (Risk of a trace). *Given a HMM* $\mathcal{M}$*, the risk of trace* $\tau \in \mathcal{L}(\mathcal{M})$ *is:*

$$R_r(\tau) := \sum_{\pi \in \Pi^h_{\mathcal{M}}} Pr(\pi \mid \tau) \cdot r(\pi_{\downarrow}).$$

[3] We use determinstic observation functions for concise definitions. Non-deterministic observation functions can be expressed via a blowup of the HMM, see e.g. [**empty citation**]

The risk of a trace can be used to partition the set of all traces.

Definition 5 (Safe traces/Unsafe traces). *Given a HMM $\mathcal{M}$, a* safe threshold $\lambda_s \in \mathbb{R}$ *and an* unsafe threshold $\lambda_u \in \mathbb{R}$ *with $\lambda_s \leq \lambda_u$. A trace $\tau \in \mathcal{L}(\mathcal{M})$ with $R(\tau) > \lambda_u$ is an* unsafe trace *(for $\mathcal{M}$ and λ_u). Conversely, trace τ is a* safe trace*, if $R(\tau) < \lambda_s$ (for $\mathcal{M}$ and λ_u).*

In the definition above, we deliberately do not require $\lambda_s = \lambda_u$. In particular, by picking $\lambda_s < \lambda_u$, some traces are neither safe nor unsafe. We write $\mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$ to denote the set of safe traces of length at most h and $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u}$ for the set of unsafe traces of length at most h. We often omit the sub/superscripts.

Example 2.

We are now ready to define correct monitors.

Definition 6 (Monitor). *A DFA $\mathcal{A}$ is a monitor for HMM $\mathcal{M}$ if the alphabet for $\mathcal{A}$ coincides with the observations in $\mathcal{M}$.*

Definition 7 (Missed/False alarms). *Given a monitor $\mathcal{A}$ for HMM $\mathcal{M}$, a horizon h, and thresholds $\lambda_s \leq \lambda_u \in \mathbb{R}$, the set of* missed alarms *is $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \setminus \mathcal{L}(\mathcal{A})$. The set of* false alarms *is $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s} \cap \mathcal{L}(\mathcal{A})$.*

We often drop the sub/superscripts. In terms of a classification problem, missed alarms are false negatives, while false alarms are false positives.

Definition 8 (Correct monitor). *Given thresholds λ_s, λ_u and horizon h, a monitor $\mathcal{A}$ for $\mathcal{M}$ is* correct *if $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset = \mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A})$.*

A correct monitor raises an alarm for all unsafe traces and for no safe trace:

Corollary 1. *A monitor $\mathcal{A}$ is correct iff $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \subseteq \mathcal{L}(\mathcal{A}) \subseteq \Sigma^* \setminus \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$.*

Problem statements. This paper tackles the following problem statements: Given HMM $\mathcal{M}$, thresholds λ_s, λ_u and horizon h,

1. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there missed alarms, i.e., is $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset$?
2. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there false alarms, i.e., is $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A}) = \emptyset$?
3. Find a correct monitor $\mathcal{A}$ for $\mathcal{M}$ w.r.t. λ_s, λ_u and h.

Problem statements 1 and 2 together allow checking whether a monitor is correct.

Remark 1. A correct monitor must exist, as $\mathbb{U}^{\leq h}$ is finite and thus regular.

We discuss our solution to the no-missed-alarms problem in Sec. 3, we adapt the solution to the no false alarm problem in Ssec. 3.4.

Example 3. We discuss the example from Fig. 3 using the correct monitor. Given the risk function assigning 1 to q_c and 0 to all other states, the traces $\tau_1 = dry, icy, icy$ and $\tau_2 = dry, icy$ have risks $^{19}/_{40}, ^{4}/_{40}$ respectively. If λ is $^{1}/_{4}$ and the horizon h is 3, τ_2 is the trace with maximum risk not in the monitor. And its risk is below λ. Thus, the monitor does not have any missed alarms.

Luko: Include verified monitor learning problem.

3 Monitor Verification

In this section we present our approach to the monitor correctness problem, which reduces checking the existence of missed alarms to a policy synthesis problem on colored MDPs. The latter is well studied. We first formalize this policy synthesis problem and then discuss the transformation in multiple steps. In our presentation, we focus on the no-missed-alarms problem with traces of exactly the length of the horizon. At the end of the section, we generalize our construction to both the no-false-alarms problem and traces of length at most the horizon.

3.1 Relating Missed Alarms with Color-Consistent Policies

Luko: There used to be (Labeled) here, but are colored MDPs ever called Labeled MDPs. Are that not just normal MDPs?

Colored MDPs are an extension to MDPs that allow expressing dependencies between states that policies must adhere to. The following definition simplifies the notion of colored MDPs from [**empty citation**], and suffices for our needs:

Definition 9 (Colored MDP). *Given an MDP $\mathcal{M}$ with states S, a* colored MDP *is a tuple $\mathcal{M}^C = (\mathcal{M}, C, c)$, where C is a set of* colors*, and $c\colon S \to C$ maps states to colors.*

Definition 10 (Color-consistent scheduler). *A memoryless scheduler σ for a colored MDP $\mathcal{M}^C$ is* color consistent[4] *if for every two states s and s', $c(s) = c(s')$ implies $\sigma(s) = \sigma(s')$.*

The set of all color-consistent schedulers is denoted Σ_c. Policy synthesis for colored MDPs asks to find a color consistent policy such that the induced reachability probability is above (or below) a certain threshold. Policy synthesis for colored MDPs is NP-hard.

Theorem 1 (Main theorem). *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, threshold λ_s, horizon h, and risk r, there exists a colored MDP $\mathcal{M}^C$ such that*

$$\exists \sigma \in \Sigma_c.\ Pr_{\sigma}^{\mathcal{M}^C}(\Diamond T) \geq \lambda_s \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}).\ \tau \in \mathsf{mA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Luko: Where does T come from? Add to the start of the theorem?

Our proof is constructive and we show that we can use the construction to find a $\tau \in \mathsf{mA}(\mathcal{A})$, whenever such a τ exists.

Outline of the proof. The proof is a direct consequence of Lemmas 1 and 2 and **??** below. Before we discuss these lemmas, we observe that on the left-hand side of Theorems 1, the monitor, horizon, observations, and risk do not occur, they must be encoded into the colored MDP. Furthermore, while missed alarms are defined using conditional probabilities, the policy synthesis problem is over reachability probabilities. We describe our transformation in several steps. In Subsection 3.2 we encode the monitor into the HMM and transform the HMM to both include the horizon and the risk. In Subsection 3.3 we resolve the conditioning on traces, while replacing observations from the HMM.

[4] Colored MDPs with color-consistent policies coincide with memoryless policies for partially observable MDPs. However, POMDPs often consider history-dependent (belief-based) policies. We use colored MDPs to avoid any confusion.

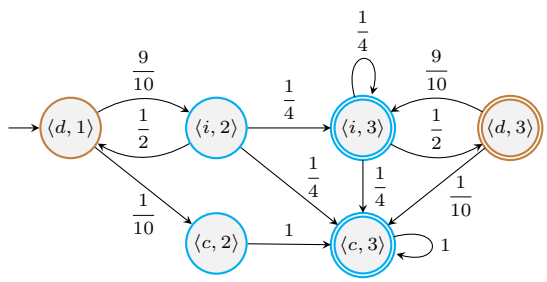


Fig. 5: The central example HMM and faulty monitor $\mathcal{B}$ transformed into the CTP problem. States are named with their HMM state, $\{d, i, c\}$ and their monitor state, $\{1, 2, 3\}$. States in the alarm states set A have been marked accepting.

3.2 The (Acyclic) Conditional Trace Probability Problem

First, we show how asking for a missed alarm can be rephrased into the conceptually simpler *conditional trace probability* (CTP) problem. We will further simplify the problem by incorporating both the horizon and risk such that we are left with a problem purely on acyclic HMMs.

CTP problem Before we combine the HMM and the monitor, we have to modify the monitor. Since we are looking for missed alarms, we are interested in all traces that are not in the monitor. We thus take the complement of the monitor. Next, we take (a mild variation of) a standard product construction [6] between HMM and DFA. We also keep track of all resulting states for which the complement monitor accepts in a set of alarms states F of the product HMM.

Example 4. Figure 5 shows the product of the HMM and monitor $\mathcal{B}$ from Figure 3. In particular, when starting in the initial states d and 1, respectively, the HMM can transition to the i state with probability $9/10$. This is an *icy* state, and thus the monitor takes the *icy* transition to the 2 state. This results in the new product state of i and 2, with a transition from $\langle d, 1\rangle$ to $\langle i, 2\rangle$ with probability $9/10$.

The set of Alarm states are all the states in monitor $\mathcal{B}$ which do not accept. Thus they are the initial $\langle d, 1\rangle$ state and any product states where the monitor is in state 3.

Definition 11 (HMM product). *Given a HMM $\mathcal{M} = (S, \iota^{\mathcal{M}}, \mathbf{P}, Z, obs, r)$ and a monitor for $\mathcal{M}$, $\mathcal{A} = (Q, \Sigma, \delta, \iota^{\mathcal{A}}, F')$. The product HMM $\mathcal{M}_{\times\mathcal{A}}$ is the HMM:*

$$\mathcal{M}_{\times\mathcal{A}} = (S \times Q, \langle \iota^{\mathcal{M}}, \iota^{\mathcal{A}}\rangle, \mathbf{P}', obs', r')$$

$$\mathbf{P}'(\langle s, q\rangle)(\langle s', q'\rangle) = \begin{cases} \mathbf{P}(s, s') & \textit{if } \delta(q, obs(s')) = q' \\ 0 & \textit{otherwise} \end{cases}$$

$$obs'(\langle s, q\rangle) = obs(s)$$

$$r'(\langle s, q\rangle) = r(s)$$

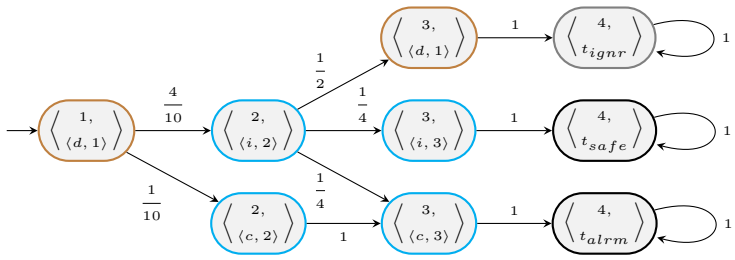


Fig. 6: ACTP, brown and cyan are dry and icy observations. Black is the new z_end observation, and gray is the new z_ignore observation. All states are named with the step i, model state s and monitor state j as $\left\langle \begin{matrix} i, \\ \langle s, j \rangle \end{matrix} \right\rangle$.

The alarm states become:

$$F = S \times F'$$

In the product, we can find a trace whose conditional trace probability exceeds a threshold iff there is a missing alarm. We state the decision problem that needs to be solved on the product HMM as it is key to our computational complexity analysis in Seciton 5.

Definition 12 (CTP Decision Problem). *Given a HMM $\mathcal{M}$ with states in S, a unary encoded horizon h, a set of alarm states $F \subseteq S$, a risk function r, and a threshold $\lambda_s \in (0, 1]$,*

$$\exists \tau \in \mathcal{L}(\mathcal{M}). \sum_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in F} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow}) \geq \lambda.$$

We denote the set of witnesses τ to a CTP decision problem as $\mathsf{CTP}(\mathcal{M}, h, F, \lambda_s)$. The following lemma states the correctness of the transformation and follows directly from the definition of missing alarms and the product with the complement.

SJ: Nevertheless, we should have a (slightly) more formal proof

Lemma 1. *Using the notation from Theorems 1, Definition 11 and Definition 12:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}_{\times \overline{\mathcal{A}}}, h, F, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{=h}_{\mathcal{M}, \lambda_s}(\mathcal{A}).$$

ACTP problem We further simplify the problem by combining the horizon, risk and the newly created alarm states set into the HMM. The resulting HMM can be seen as the model for the acyclic CTP problem.

Example 5. The ACTP transformation applied to the HMM from Fig. 5 can be seen in Fig. 6. We unroll the CTP HMM for a horizon of 3. Thus, the initial state becomes the tuple of step 1 and the initial state from the CTP HMM, $\langle d, 1 \rangle$. Any

transition in the model increases the step value by 1, $\langle 1, \langle d, 1\rangle\rangle$ transitions to $\langle 2, \langle c, 2\rangle\rangle$. In state $\langle 3, \langle i, 3\rangle\rangle$ we are at the horizon, and the CTP part of the state is part of the alarm states. Thus, the monitor wants to raise an alarm here. We now transition with probability 1 to the $\langle 4, t_{safe}\rangle$ state and probability 0 to the $\langle 4, t_{alrm}\rangle$ state, since the normalized risk of d is 0. The horizon state $\langle 3, \langle d, 1\rangle\rangle$ is not in A, thus we transition to the $\langle 4, t_{ignr}\rangle$ state.

Definition 13 (Unrolling with risk). *The (risk-)unrolled HMM $\mathcal{M}_{\blacktriangleright h}$ of a HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ with horizon h, risk r and alarm states $F \subseteq S$ is the HMM $\mathcal{M}_{\blacktriangleright h} = (\{1, \ldots, h\} \times S \cup \{\langle h+1, t_{alrm}\rangle, \langle h+1, t_{safe}\rangle, \langle h+1, t_{ignr}\rangle\}, (1, \iota), \mathbf{P}', Z \cup \{z_{end}, z_{ignore}\}, obs')$, with the normalized risk function $r'(s) = {}^{r(s)}/_{\max_{s \in S} r(s)}$ and new transition and observation function:*

$$\forall_{i \in \{1,\ldots,h-1\}} \quad \mathbf{P}'(\langle i, s\rangle)(\langle i+1, q\rangle) = \mathbf{P}(s)(q)$$

$$\mathbf{P}'(\langle h, s\rangle)\,(\langle h+1, t\rangle) = \begin{cases} r'(s) & \text{if } s \in F \wedge t = t_{alrm} \\ 1 - r'(s) & \text{if } s \in F \wedge t = t_{safe} \\ 1 & \text{if } s \notin F \wedge t = t_{ignr} \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbf{P}'\,(\langle h+1, t\rangle)\,(\langle h+1, t\rangle) = 1$$

$$obs'(\langle i, s\rangle) = obs(s)$$

$$obs'(\langle h, t\rangle) = \begin{cases} z_{end} & \text{if } t \in \{t_{alrm}, t_{safe}\} \\ z_{ignore} & \text{if } t = t_{ignr} \end{cases}$$

The unrolled transition relation $\mathbf{P}'$ is defined as expected. At the horizon we transition to the three new states t_{alrm}, t_{safe} and t_{ignr}[5] according to the risk.

Lemma 2. *Given an HMM $\mathcal{M}$, horizon h, alarm states F and threshold λ:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}, h, F, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}).\ \sum_{\pi \in \Pi^{\mathcal{M}}} Pr^{\mathcal{M}}(\pi s_a \mid \tau \cdot z_{\mathsf{end}}) \geq \lambda,$$

where z_{end} is the final observation and s_a is the alarm state.

SJ: I wonder whether we should simplify notation on the right hand side. I am not too fond of the z and s_a symbols.

The proof of the lemma essentially starts by using that the lhs of the lemma is by definition $\exists \tau \in \mathcal{L}(\mathcal{M}).\ \sum_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in F} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow}) \geq \lambda.$

SJ: keep this in sync with def Def. 12

Luko: I am not sure about either s_a or *alarm*, the first is shorter, but the second is a lot more descriptive

SJ: Put a slightly longer proof in the appendix

3.3 Reduction to Consistent Policy Synthesis

For Lemma 2, we must find a trace such that a conditional reachability probability exceeds a threshold. We reformulate this into a policy synthesis problem in MDPs.

[5] We add an $\langle h+1, t_{ignr}\rangle$ state, instead of redirecting all traces we don't care about to the $\langle h+1, t_{safe}\rangle$ state, to more easily modify the transformation for the no-false-alarms problem.

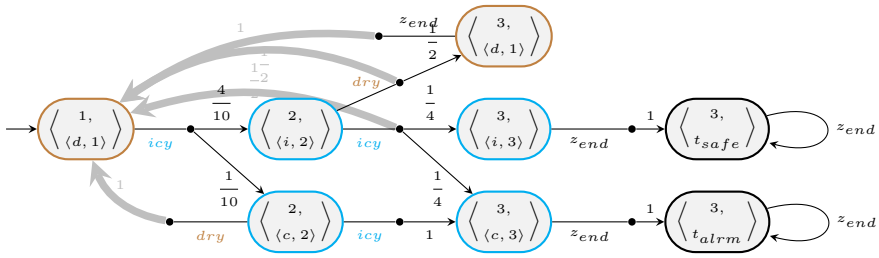


Fig. 7: The central example transformed into an mdp to be used as the model in the MDP policy synthesis problem. For conciseness, all unreachable states are removed. Also any actions which return for all states in the same step are removed.

The transformation combines two ideas: First, in every state, the policy can select the next observation, inspired by [**empty citation**]. Second, we reformulate conditional reachability probabilities by (standard) regular reachability probabilities, using a construction in [**empty citation**].

Example 6. The consistent policy synthesis transformation applied to the ACTP HMM in Fig. 6 can be seen in Fig. 7. The transformation applied to state $\langle 2, \langle i, 2\rangle\rangle$ in the ACTP HMM results in the following. The three transitions are split into the one transition going to a *dry* state, $\langle 3, \langle d, 1\rangle\rangle$, and two transitions going to the two *icy* states in step 3. We plate the first transition behind the *dry* action. The other two transitions are placed behind the *icy* action. Both of the action have a total probability of $\frac{1}{2}$, thus a return transition is added to the initial state of $\frac{1}{2}$ probability to both actions.

State $\langle 2, \langle c, 2\rangle\rangle$ does not reach any *dry* states, we still add a *dry* action redirecting to the initial state, since we want all states with the same step value to have the same actions.

State $\langle 3, \langle d, 1\rangle\rangle$ only has a transition to an z_{ignore} state. We fully remove this state and the observation. Since it does not have an z_{end} action, while the other 3 step states do. The state gets an z_{end} action to the initial state.

Definition 14. *Given an acyclic HMM $\mathcal{M}_{\blacktriangleright h} = (S, \iota, \mathbf{P}, Z, obs, r)$ as in Definition 13, we define the colored MDP $\mathcal{M}_{\gtrdot h} = ((S', \iota, Act, \mathbf{P}'), C, c)$ with $Act = Z \setminus \{z_{ignore}\}$,*

$$\mathbf{P}'(\langle i, s\rangle, z)(\langle j, q\rangle) := \begin{cases} \mathbf{P}'(\langle i, s\rangle)(\langle j, q\rangle) & \text{if } obs(\langle j, q\rangle) = z \\ \displaystyle\sum_{(\langle k,q'\rangle)\in S' \mid obs(\langle k,q'\rangle)\neq z} \mathbf{P}'(\langle i, s\rangle)(\langle k, q'\rangle) & \text{if } (\langle j, q\rangle) = \iota \\ 0 & \text{otherwise} \end{cases}$$

$C = \{1, \ldots, h\}$ and c is given such that $c(\langle i, s\rangle) = i$.

Thus we transition normally to a state if the action and observation of the target state are equal, otherwise we set the transition probability to zero. All the remaining probability mass is redirected towards the initial state.

The above construction allows for conditioning on a trace τ by constructing a policy σ that selects the ith observation of τ in the state with step i.

Definition 15 (Trace consistent policy). *Given two sets S and Z and an MDP $\mathcal{M}$ with states in $\mathbb{N} \times S$, and actions in $Z \cup \{end\}$. Together with a trace $\tau \in Z^\star$. A* trace consistent policy *is of the form*

$$\sigma_\tau(\langle i, s\rangle) = \begin{cases} \tau^{(i)} & \text{if } i \leq |\tau|, \\ end & \text{otherwise.} \end{cases}$$

Lemma 3.

$$\exists \sigma \in \Sigma_c . Pr_\sigma^{\mathcal{M}^C}(\Diamond T) \geq \lambda \quad \text{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}). \sum_{\pi \in \Pi^{\mathcal{M}}} Pr^{\mathcal{M}}(\pi s_a \mid \tau \cdot z_{\mathsf{end}}) \geq \lambda,$$

In the ACTP HMM we only have the observations and conditional probability on a trace left. To incorporate this into the model we will use one novel idea. Instead of transitioning to a state and observing it, we will allow the model to choose what observation to observe. We thus move away from HMMs and into MDPs. This has two results. Firstly, instead of talking about sequences of observations, instead we will consider sequences of actions taken. Secondly, any trace can be transformed into a policy of the MDP. We will first discuss how to create this MDP. Afterward, we show how to ensure how, traces and policies can be made equivalent by coloring the MDP.

Corollary 2. *There exists a map $t(\sigma_\tau) = \tau$ which, given a trace consistent policy σ_τ finds its associated trace τ.*

However, when finding the maximal trace, we might not find a trace consistent policy. To fix this we color the MDP such that color-consistent policies coincide with trace consistent policies on our MDPs. As colors Given that all *end* states sinks, color consistent policies with the trace consistent policies.

Remark 2. All states gain all actions in $\mathbf{P}''$, we can later prune any actions for all states S^C of a color C when these actions only redirect to the initial state from S^C.

Now finding a trace with the conditional trace probability above λ_s coincides with finding a color consistent policy such that

$$Pr^\sigma(\Diamond \langle h+1, t_{alrm}\rangle) > \lambda_s$$

3.4 Adapting to No-False-Alarms and Smaller Traces

- Add end transitions to non horizon states.
- We want to minimize alarm probability, not possible.
- Instead maximize safe probability.

Traces of Length up to the Horizon The aproach for Thm. 1 works only for traces of exactly the length of the horizon.

Luko: Something about it being hard for ACTP but easy for Pol Syn

We transform Def. 14 such that the following theorem holds:

Theorem 2. *Luko: Something relating the modified transformation to missing alarms on $\leq h$*

Proof. We modify the transformation from Thm. 1 in the policy synthesis step. We add a new initial state, ι', it gets a seperate coloring from all other states. This state contains an action for each possible length l of a trace up to the horizon. The actions now transition to the state $\langle l, \iota \rangle$.

$$\mathbf{P}'(\iota', l)(\langle l, \iota \rangle) = 1$$

We now show this transformation is correct.

In order to verify that there are no-missed-alarms for all $l < h$, we could use the transformation from Thm. 1 with the horizon equal to all $l < h$. This would neccesitate doing policy synthesis for h colored MDPs.

$$\exists_\tau \exists_{l \leq h} Pr^{\sigma_\tau}[\Diamond alarm] > \lambda_s$$

If any of these h policy synthesis problems can find a policy σ_τ, there exists a trace $\tau \in \mathsf{mA}_{\leq h}$.

We combine these h colored MDPs into one colored MDP in the following way. We add a new initial state, and give it h actions $\{1, \ldots, h\}$. Action $l \in \{1, \ldots, h\}$ points, with probability 1, to the initial state of colored MDP $\mathcal{M}_{\geqslant l}$. This is equivalent to solving policy synthesis on the h individual MDPs.

Luko: Maybe note that the $\mathcal{M}_{\geqslant l}$ only differed when they where in the CTP stage by the horizon, nothing else.

We now note that for any $l < l' \leq h$ the initial state $\langle 1, \iota^{CTP} \rangle$ of $\mathcal{M}_{\geqslant l}$ is bisimilar to the state $\langle l' - l + 1, \iota^C TP \rangle$ in $\mathcal{M}_{\geqslant l'}$. We now claim that the result of bisimulation minimization on this combined colored MDP is described by the transformation described at the start of the proof.

Example 7. The transformation in Thm. 2 can be seen in Fig. 8

No-False-Alarms Problem We modify the transformation from Thm. 1 such that it solves the no-false-alarms problem. The no-false-alarms problem differs in two ways from the no-missed-alarms problem. Fristly, instead of finding a trace not in the monitor, we have to find a trace in the monitor. Secondly, this trace should have a conditional probability below the threshold. We solve these one at a time after stating Thm. 1 for no-false-alarms.

Theorem 3. *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, threshold λ_u, horizon h, and risk r, there exists a colored MDP $\mathcal{M}^C$ and a λ such that*

$$\exists \sigma \in \Sigma_c . Pr_\sigma^{\mathcal{M}^C}(\Diamond T) \leq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{fA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Proof. We reuse the transformation from Thm. 1 with the following modifications. We no longer take the complement of the monitor. Instead of doing reachability on the alarm state, we instead do reachability on the safe state.

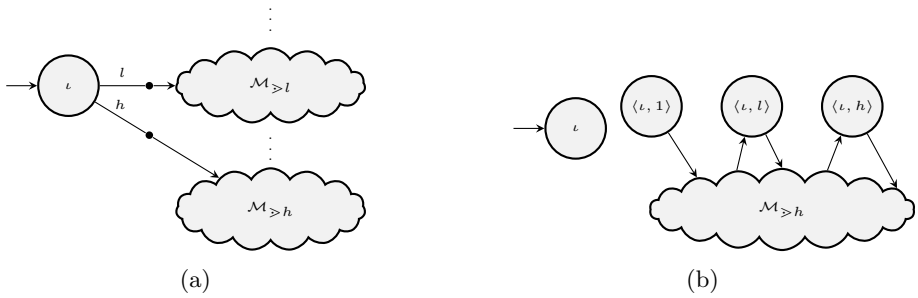


Fig. 8

4 Learning Monitors

In this section we aim to learn correct monitors. We use the automata learning framework based on a Minimally Adequate Teacher (MAT) [4]. To ensure that the monitors are correct, we use the verification algorithm outlined above. We furthermore discuss some improvements that bridge the gap between common assumptions in MAT automata learning and our monitor setup.

MAT framework We briefly recap the MAT framework, for details see [12]. A minimally adequate teacher is a teacher answer two types of questions: A membership query, which in our setup means *should a trace be part of the monitor?* and an equivalence query, *is this monitor correct?* Furthermore, if the answer to an equivalence query is negative, we must provide a counterexample that witnesses why the monitor is not correct. Various algorithms implementing the MAT framework for DFA learning exist. For the purpose of this paper, we use the $L^\star$ algorithm [4]. Roughly, the learner will ask the teacher membership queries to construct a hypothesis, i.e., some monitor that is consistent with the answers from the teacher. Once such a hypothesis is constructed, the monitor is verified using an equivalence query.

Verification as a MAT When learning monitors with a MAT we will provide the HMM $\mathcal{M}$, a risk function r, a horizon h, a learning threshold λ_l, and the safe and unsafe thresholds λ_s and λ_u. After learning we obtain the monitor $\mathcal{A}$.

We now define the MAT framework for learning monitors by defining the MQ and the EQ. We define the EQ by adapting Def. 5 for $\tau \in Z^\star$.

Definition 16. *Given any trace $\tau \in Z^\star$, and a horizon h, a MQ_{λ_l}, is a function f, where $f(\tau)$ is unsafe iff $\tau \in \mathcal{L}(\mathcal{M})$[6] and $\tau \in \mathbb{U}_{\lambda_l}^{\leq h}$.*

[6] Defining a trace out of the HMM safe is an arbitrary decision, as they do not matter for correctness. However, in the MAT framework a trace always has to be either safe or unsafe

Such a function can be defined by keeping track of the probability of being in each state after every observation from the trace [9].

We adapt the verification described in Sec. 3 to define the EQ.

Definition 17. *Given a HMM $\mathcal{M}$, and a monitor $\mathcal{A}$, an EQ_{λ_s,λ_u} is a function g such that $g^{\mathcal{M}}(\mathcal{A})$ holds if $\mathcal{A}$ is correct for $\mathcal{M}$ with λ_s, and λ_u (in the sense of Def. 8). And returns a counter example trace if $\mathcal{A}$ is not correct.*

The EQ requires checking both for no-missing-alarms, and for no-false-alarms. Each check follows the steps as described in Sec. 3. We first apply the transformation in order to create a colored MDP problem. Which is then solved using PAYNT. If a false alarm or missed alarm trace is found it is returned as a counter example.

Lemma 4. *Given a MAT with an EQ_{λ_s,λ_u} and an MQ_{λ_l}, a monitor learned with $L\star$ is correct as long as $\lambda_u \leq \lambda_l \leq \lambda_s$.*

Proof. If, while learning a monitor $\mathcal{A}$ a trace τ is deemed safe by MQ_{λ_l} it cannot be given as a counterexample by $\mathrm{EQ}_{\lambda_s,\lambda_u}$ on $\mathcal{A}$, since $\lambda_u \leq \lambda_l \leq \lambda_s$, and $\tau \notin \mathcal{A}$ by $\mathrm{L}^\star$. Similarly, MQ_{λ_l} and $\mathrm{EQ}_{\lambda_s,\lambda_u}$ also agree on unsafe traces. Now, by correctness of $\mathrm{L}^\star$, a learned monitor $\mathcal{A}$ has to hold for $\mathrm{EQ}_{\lambda_s,\lambda_u}$, and thus be correct.

Note that we do not have to set λ_s and λ_u to the learning threshold λ_l. This allows the EQ to have a gray area given by the interval (λ_s, λ_u). Traces which have a risk in this interval can be in or out of the monitor, while it is still correct. We investigate the results of this gray area in Sec. 6 **Q4**.

Improvements. An alternative to the EQ in Def. 17 is a conformance query [**empty citation**], where we test a monitor using a large amount of sampling. In our approach we use a hybrid of the twe queries. Monitors produced early in the learning process often contain many missed alarms and false alarms. Verification can find them, however, it has a constant upfront cost. Conformance queries can often find a counter example faster if one has a high probabilty of occuring.

5 Computational Complexity

Theorem 4. *Checking whether a monitor is correct given a unary encoded horizon is coNP-complete.*

In fact, we study in this section the dual to this problem, i.e., checking the existence of a counterexample. We call this problem *monitor co-verification*. For monitor co-verification, *membership* in NP follows as false alarms or a missed alarm (of length up to horizon) are witnesses. Verifying whether a trace is indeed a false alarm can be done in polynomial time by first checking whether the automaton accepts it and then computing the trace risk and comparing it to the thresholds. Each of these steps can be done in polynomial time [**empty citation**].

SJ: Maybe this result is so crucial that we should explain somewhere how this works in polynomial time. I guess it also follows from the construciton in section 3 if we restrict the monitor to just contain one trace?

To establish NP-hardness, we consider the CTP problem from Definition 12. As a solution to the monitor co-verification problem suffices to solve the CTP problem (using a trivial monitor), this implies NP hardness of the former.

Lemma 5. *The CTP Decision Problem is NP-hard.*

The lemma holds even for $\lambda = 1$ and acyclic HMMs. We reduce the problem from 3CNF. We outline the construction below.

Consider a 3CNF formula $\varphi = \bigwedge c_1 \dots c_m$ over variables X, $|X| = n$, with clause $c_j = \ell_j^1 \vee \ell_j^2 \vee \ell_j^3$ and each literal $\ell_j^i \in \{x, \neg x \mid x \in X\}$. We transform this into a CTP instance with $\lambda = 1$ and an acyclic HMM $\mathcal{M}_\varphi$ with observations $Z = \{\#, \bot, \top\}$. The only state with positive risk is a dedicated state t with $r(t) = 1$. The crux of the construction is that there is a trace with risk 1 iff there is a satisfying assignment to φ. In the constructed HMM, there is a trace with risk 1 iff there is a trace where all corresponding paths end in state t.

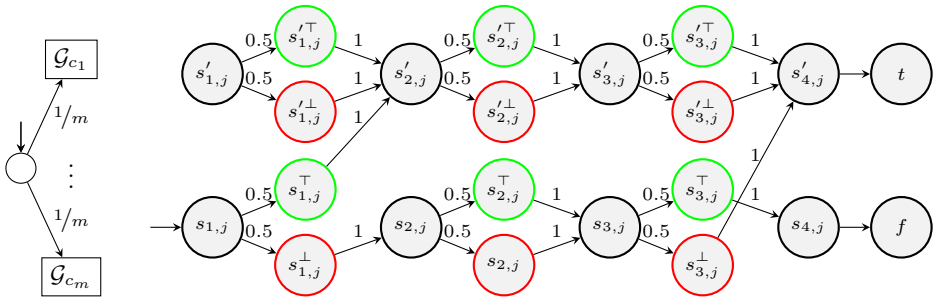
SJ: make sure that corresponding paths are the right formal notion

Before we give a formal definition of $\mathcal{M}_\varphi$, we give some intuition. We represent assignments $\alpha \colon X \to \{\bot, \top\}$ by traces through the HMM $\#\#\cdot\alpha(x_0)\cdot\#\cdot\alpha(x_1)\cdot\dots\cdot\#$. We create gadgets for every clause. The gadgets are connected as in Figure 9a: That is, intuitively, $\mathcal{M}_\varphi$ randomly selects a clause c_j with probability $1/m$ and transitions into gadget $\mathcal{G}_j$ defined below. Next, we show that the gadget reaches positive risk in every gadget only if the corresponding clause is satisfied by the assignment.

The gadget $\mathcal{G}_j$ intuitively 'evaluates' c_j with respect to an assignment α, by starting in $s_{1,j}$ with a gadget for this clause, as exemplified in Fig. 9b. We enter the gadget with trace $\# \cdot \alpha(x_0) \cdot \# \cdot \alpha(x_1) \cdot \dots \cdot \#$, i.e., the first observation has been matched by the initial state. A path (or its trace) through $\mathcal{G}_j$ 'reads' variable x_i in state $s_{i,j}$ and transitions to $s_{i,j}^\top$ or to $s_{i,j}^\bot$. However, only one of these paths corresponds to the trace. Thus, in every gadget, there is only one path that corresponds to a given trace and so they can be used interchangeably. State $s_{i,j}$, $s_{i,j}^\top$ and $s_{i,j}^\bot$ have observations $\#, \top, \bot$, respectively. We note that a path reaches $s_{i,j}^\top$ if $\alpha(x_i) = \top$ and (analogously for $\bot$). From there onwards, if the literal x_i (or $\neg x_i$) occurs in clause c_j, we transition from $s_{i,j}^\top$ (or $s_{i,j}^\bot$, resp.) to $s'_{i+1,j}$, otherwise, we transition to $s_{i+1,j}$. A path ends in t only via a state $s'_{i,j}$, which is only possible if the path visits a state corresponding to a literal in the clause. Thus, (conditionally) reaching state $s'_{i,j}$ with positive conditional probability means that the partial assignment $\alpha(x_1), \dots, \alpha(x_i)$ satisfies clause c_j, while reaching $s_{i,j}$ with positive conditional probability means that the clause is either unresolved or unsatisfied given the partial assignment.

Formally, we construct the HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ and set $\lambda = 1$:

- $S = \{s_{i,j}, s_{i,j}^\bot, s_{i,j}^\top, s'_{i,j}, s'^{\bot}_{i,j}, s'^{\top}_{i,j} \mid i \in \{1, \dots, n+1\}, j \in \{1, \dots, m\}\} \cup \{s_\iota\}$.
- $\mathbf{P}$ is given such that for all $i \in \{1, \dots, n\}, j \in \{1, \dots, m\}$:
  - $\mathbf{P}(s_\iota, s_{1,j}) = \dfrac{1}{m}$,
  - $\mathbf{P}(s_{i,j}, s_{i,j}^\top) = \mathbf{P}(s_{i,j}, s_{i,j}^\bot) = \dfrac{1}{2} = \mathbf{P}(s'_{i,j}, s'^{\top}_{i,j}) = \mathbf{P}(s'_{i,j}, s'^{\bot}_{i,j})$,
  - $\mathbf{P}(s_{i,j}^\top, s'_{i+1,j}) = [x_i = \ell_j^k$ for some k$]$, $\mathbf{P}(s_{i,j}^\top, s_{i+1,j}) = [x_i \neq \ell_j^k$ for all k$]$,
  - $\mathbf{P}(s_{i,j}^\bot, s'_{i+1,j}) = [\neg x_i = \ell_j^k$ for some k$]$, $\mathbf{P}(s_{i,j}^\bot, s_{i+1,j}) = [\neg x_i \neq \ell_j^k$ for all k$]$
  - $\mathbf{P}(s'_{n+1,j}, t) = 1 = \mathbf{P}(s_{n+1,j}, f)$
- $Z = \{\top, \bot, \#\}$ and $obs(s_{i,j}^\top) = (s'^{\top}_{i,j}) = \top$, $obs(s_{i,j}^\bot) = (s'^{\bot}_{i,j}) = \bot$, $obs(s_{i,j}) = obs(s'_{i,j}) = \# = obs(s_\iota) = obs(t) = obs(f)$.



(a) HMM. (b) Gadget demonstrated for $x_1 \vee \neg x_3$

Fig. 9: Illustrations for Lemma 5, with m clauses and variables x_1, x_2, x_3.

– $r(t) = 1$ and $r(s) = 0$ for all $s \in S \setminus \{t\}$.

The construction runs in polynomial time. The formal proof of its correctness follows the explanation above precisely.

Approximability Approximating the verification of monitors is also hard, in the sense that it is hard to approximate the maximal risk that a monitor admits.

The maximal risk problem is an optimization problem

Definition 18 (CTP Optimization Problem). *Given an HMM $\mathcal{M}$ with states in S, a unary encoded horizon h, a set of alarm states $A \subseteq S$:*

$$\max_{\tau \in \mathcal{L}(\mathcal{M})} \sum_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in A} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow})$$

SJ: check notation for paths of length h

SJ: Sync with CTP definition

Lemma 6. *The CTP Optimization Problem is APX-hard.*

This follows from a strict reduction from MAX-3SAT, which is an inapproximable and APX-hard problem [7]. The construction coincides with the reduction in Lemma 5 by observing that the conditional probability to reach a t state is given by $1/m$ times the number of satisfied clauses, i.e., we can compute the maximal number of satisfied clauses in φ on the HMM $\mathcal{M}_{\varphi}$.

6 Experiments

In this section, we empirically evaluate the monitor verification (Sec. 3) and monitor learning (Sec. 4) using our prototype implementation called ToVer. We aim to answer the following questions: We evaluate the verifier with the following questions:

Q1 What is the scalability of the monitor verification approach?
Q2 Which steps in the monitor verification are most expensive?

We evaluate the learning algorithm using the following questions:

Q3 How does a verification-based oracle compare to using conformance checking?
Q4 What is the effect of safe and unsafe thresholds?

Setup. The ToVer tool is built in Python and C++ on top of the Storm model checker [8] for most data structures and for the filtering (Membership queries in Sec. 4) [9]. We use PAYNT [3] to verify the generated colored MDP (Def. 9). The learner uses the AAlpy automata learning framework [10]. All experiments are run using exact arithmetic on a single thread of an AMD Ryzen TRP 5965WX and with a memory limit of 15 GiB.

Benchmarks. We combine benchmarks from the literature with some novel benchmarks. We take the Aiport, Refuel, Evade, and Hidden-Incentive benchmark families from [9]. We add Icy-Driving, based on the central example but with more states describing how close the car is to crashing and SnL, a benchmark based on the game "Snakes and Ladders", with a 10x10 game board and one player. While the existing benchmarks contain many observations, the new benchmarks only have a few different observations. Every benchmark is analyzed with respect to different horizons and in different sizes. The benchmarks have up to hundreds of states and thousands of transitions. Details are given in appendix A.

Q1 and Q2: Efficiency and Scalability of Monitor Verification

Setup. We run the ToVer verifier on the benchmarks as described above. We obtain monitors from the ToVer learning experiments with no gray area. Every version of the benchmark is run on the first monitor which got through the conformance checking, a monitor from around the middle of the learning, and the final correct monitor obtained through learning. For each monitor we verify $\lambda_s = \lambda_u = 0.3$, and find the minimum in monitor risk and maximum out of monitor risk. Each run is named according to the benchmark and a sequential counter e.g., A-10 for the tenth Airport run.

How scalable is verification? We are in general able to verify HMMs with hundreds of states and above a thousand transitions, and monitors with hundreds of states and tens of thousands of transitions. We now highlight some general trends using table 1, which is a subset of appendix B. Worse monitors, A-0 to A-3 from Airport, are easier to find an incorrect trace for than good monitors, A-8 to A-11. However, verifying an extreme bound can still be hard, as seen with A-2 compared to A-3. Also, missed alarms seem harder to find than false alarms. Increasing the horizon can lead to very high runtime even for very simple models, as seen in I-14/15 compared to I-34/35, form Icy-Driving. Lastly, we are able to verify HMMs and Monitors with large numbers of states as seen in E-20 to E-23 of Evade.

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
AirportA-3	A-0	3/10	10	MA	45	113	18	25	450	0.92	0.31	0.41	0.94
AirportA-3	A-1	-	10	MA	45	113	18	25	450	6.24	0.31	5.73	1.00
AirportA-3	A-2	3/10	10	FA	45	113	18	25	450	0.95	0.31	0.43	0.08
AirportA-3	A-3	-	10	FA	45	113	18	25	450	1014.53	0.31	1014.02	0.01
AirportA-3	A-8	3/10	10	MA	45	113	18	128	2304	833.83	0.56	832.93	-
AirportA-3	A-9	-	10	MA	45	113	18	128	2304	788.22	0.56	787.31	0.30
AirportA-3	A-10	3/10	10	FA	45	113	18	128	2304	1143.18	0.53	1142.35	-
AirportA-3	A-11	-	10	FA	45	113	18	128	2304	1106.29	0.55	1105.43	0.30
Evade	E-20	3/10	9	MA	385	1473	325	328	106600	49.11	33.35	8.23	-
Evade	E-21	-	9	MA	385	1473	325	328	106600	50.67	33.39	8.20	0.27
Evade	E-22	3/10	9	FA	385	1473	325	328	106600	49.75	34.26	8.14	-
Evade	E-23	-	9	FA	385	1473	325	328	106600	50.18	33.36	7.88	0.30
Icy-Driving	I-34	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-35	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33
Icy-Driving	I-14	3/10	25	FA	3	6	2	2	4	1225.78	0.01	1225.76	-
Icy-Driving	I-15	-	25	FA	3	6	2	2	4	1173.95	0.01	1173.94	0.33

Table 1: Subset of verification experiment results found in Appendix B. Dashes in the λ_l column indicate finding the extreme instead of verifying a bound. Dashes in λ^{found} indicate no trace was found with a risk above (or below, for FA) the indicated threshold.

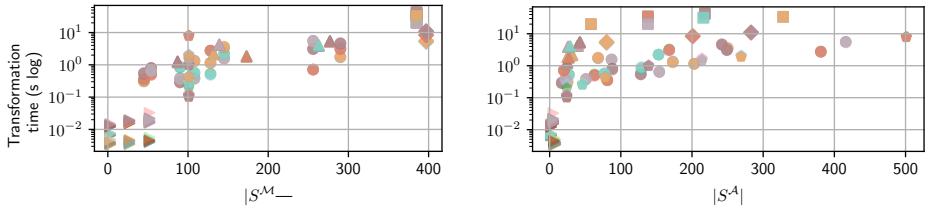
What steps are most expensive? The time spent to verify a monitor can mostly be split up in time to apply the transformation from Sec. 3, and time spent to verify the created colored MDP using PAYNT. Verifying a colored MDP is an NP-complete problem, and its runtime is heavily dependent on the internal structures and thresholds used. From table 1, the evade benchmarks are very large but take only around eight seconds to verify by PAYNT, where A-10 and A-11 are much smaller, but take around a thousand seconds to verify. The transformation step does seem to increase if the size of the HMM or the monitor increases as can be seen in Fig. 10. But this again does differ between benchmarks.

Q3: Efficiency of Monitor Learning

Setup. We use the MAT framework from Sec. 4 and configure the queries as follows. We set the learning threshold $\lambda_l = 0.3$, and we run the ToVer learner with $\lambda_s = 0.35, \lambda_u = 0.1$ and with $\lambda_u = \lambda_l = \lambda_s = 0.3$. The role of $\lambda_s \neq \lambda_u$ is evaluated in **Q4** below. [7] Before executing an EQ, we run conformance checking with 100 samples for λ_s and 100 samples for λ_u (see Sec. 4). As a baseline, we do not use an EQ oracle, but rather only use a conformance query as described in Sec. 4. This conformance query decides a monitor is correct after sampling 100000 traces[8]. We only discuss the results of learning with $\lambda_u = \lambda_l = \lambda_s$ and with $\lambda_u \leq \lambda_l \leq \lambda_s$ together for **Q3**. In **Q4** we discuss the impact of setting $\lambda_u \neq \lambda_s$.

[7] The role of λ_l was evaluated with sampling as can be seen in Appendix E.

[8] We investigate the role of the number of samples in Appendix Ec



(a) Transformation time steps against HMM states.

(b) Transformation time steps against monitor states.

Fig. 10: Time in the transformation step compared to the size of the HMM $\mathcal{M}$ or the monitor $\mathcal{A}$ for ToVer verification.

(a) ToVer, $\lambda_u = \lambda_l = \lambda_s$

(b) ToVer, $\lambda_u < \lambda_l < \lambda_s$

(c) Baseline

Fig. 11: $\mathbf{Q}_3$ All monitor correct: Actual alarm and actual no-alarm thresholds from monitors learned with ToVer learning and baseline learning. The line between the green and gray area is λ_u, the line between the red and gray area is λ_s. The dotted line is λ_l. Any benchmarks which did not finish are left empty.

Are the monitors correct? First, we investigate the *actual* performance of the learned monitors. That is, for any monitor we can compute the trace in the monitor that has the lowest risk (actual alarm threshold, $\lambda_u^{\min}$) and the trace not in the monitor that has the highest risk (actual no-alarm threshold, $\lambda_s^{\max}$). In a correct monitor, we have $\lambda_u^{\min} \geq \lambda_u$ and $\lambda_s^{\max} \leq \lambda_s$.

SJ: is this simple to do?

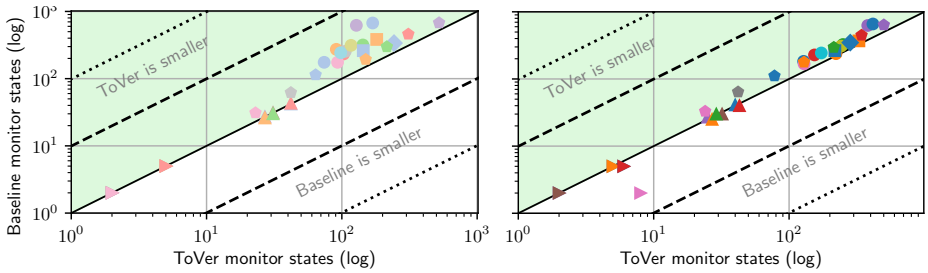
Figures 11a and 11b together and Fig. 11c show $\lambda_u^{\min}$ and $\lambda_s^{\max}$ for ToVer learning and baseline learning respectively. A monitor is correct if its red bar never touches the green area, and the green bar never touches the red area. Several incorrect monitors exist in Fig. 11c, 6 monitors learned with the baseline learner still had a missed alarm. No monitors had false alarms.

Luko: Yes, stopping the maximalization of the policy when we reach the threshold, we just continue and report the found value afterwards, intstead of just returing true/false.

How big are the monitors? For our benchmarks we are learning monitors of hundreds of states and tens of thousands transitions, as can be seen in Figs. 12a and 12b and Appendix C. For the literature on AAL, these are large automata[13, 1, 11]. Comparing the sizes of the monitors learned using ToVer and the baseline, almost no monitors are smaller when using the baseline. When using ToVer learning, monitors can be up to around 5 times smaller, but most are between 1.5 and 2 times smaller.

Luko: Mention we are learning stuff that is hard for AAL people to learn.

How fast do we learn the monitors? We describe the runtime of the ToVer learning and baseline learning in the Figs. 13a to 13c. Figures 13a and 13b show how the



(a) ToVer with $\lambda_u < \lambda_l < \lambda_s$ (b) ToVer with $\lambda_u = \lambda_l = \lambda_s$

Fig. 12: Comparison of the amount of states in monitors learned using ToVer learning compared to baseline learning.

Luko: check number

Luko: check

Luko: also check

Luko: I stopped them manually after around a day, so no real timeout. Is this enough in that case?

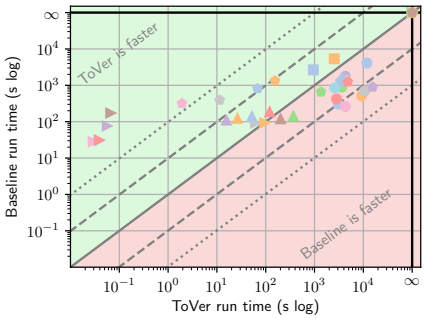
runtime of ToVer learning and baseline learning compare. Baseline takes at least about 100 seconds, while ToVer learning can take only 0.02 seconds to find a monitor both algorithms take at most around 9 hours. Several benchmarks never finished after more than 24 hours and were stopped. Neither of the two learning algorithms is clearly faster than the other.

Figure 13c displays the amount of time spent in each part of the algorithm during learning using ToVer. Most benchmarks are heavily dependent on PAYNT for the runtime. However, several Evade benchmarks spent most of their time in L$\star$. We conjecture this happens because finding counter examples is very fast for this model.

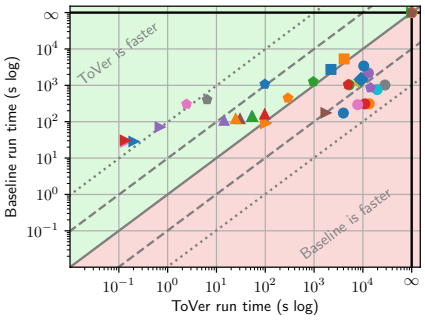
Q4: The role of λ_l, λ_u and λ_s We make use of the same setup as **Q3**, and compare between $\lambda_u = \lambda_l = \lambda_s$, without a gray area, and $\lambda_u < \lambda_l < \lambda_s$, with a gray area. This does not have an effect on the baseline learner as it only takes λ_l into account.

Figures 12a and 12b contain the size of the monitors found using ToVer learning and baseline learning. Fig. 12a contains a comparison of monitors learned with a gray area, while Fig. 12b compares ToVer learning without a gray area. Both axes are on a logarithmic scale. Most monitors are between the same size and 10 times smaller when using ToVer learning with a gray area than when using baseline learning. When not learning with a gray area the difference is smaller. Two monitors are smaller when using baseline learning, one monitor was reduced from 4 to 3 states, the other from 7 to 1 state.

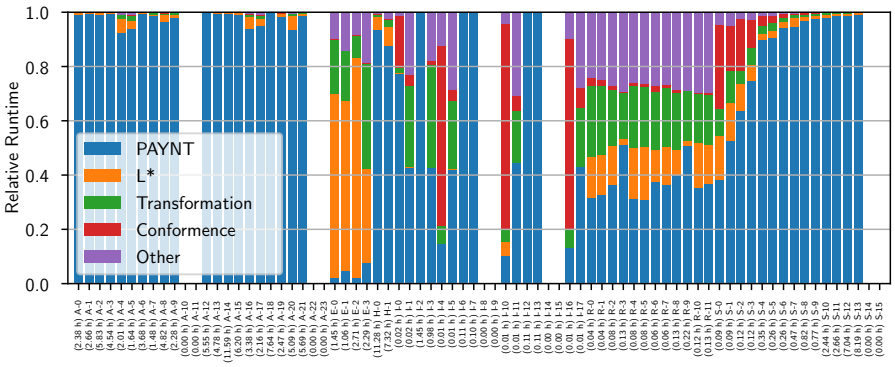
Figures 13a and 13b contain the runtime of ToVer learning compared to the baseline, when running with and without a gray area. The Icy-Driving (▶), Evade (■), and Airport (●), are generally faster when run with a gray area. In other benchmarks, adding a gray area has little effect on the runtime. A few HMMs were much faster to learn when using a gray area, one of these is the brown Icy-Driving HMM.



(a) Runtime comparison, $\lambda_u < \lambda_l < \lambda_s$



(b) Runtime comparison, $\lambda_u = \lambda_l = \lambda_s$



(c) Runtime division over steps in ToVer learning. Empty bars are HMMs for which learning did not finish.

Fig. 13: Learning experiment runtime results. In Figs. 13a and 13b points with the same symbol and similar color are the same benchmark with and without a gray area.

7 Related

- LTLf
- Predictors or so by Ziemek et al.
- MDPs as Distribution transformers
- Computing conditional probabilities (About reachability conditionals, not ending conditionals)

8 Conclusion and Future Work

SJ: Something about the size of monitor and model together

- Long traces (complexity and practical implementation).
- Unbounded traces (It woudl be great if we can add why this is hard)

References

[1] Bernhard K. Aichernig, Martin Tappler, and Felix Wallner. "Benchmarking Combinations of Learning and Testing Algorithms for Automata Learning". In: *Form. Asp. Comput.* 36.1 (Mar. 20, 2024), 3:1–3:37. ISSN: 0934-5043. DOI: 10.1145/3605360. URL: https://dl.acm.org/doi/10.1145/3605360 (visited on 01/23/2025).

[2] Roman Andriushchenko et al. "Inductive synthesis of finite-state controllers for POMDPs". In: *UAI*. Vol. 180. Proceedings of Machine Learning Research. PMLR, 2022, pp. 85–95.

[3] Roman Andriushchenko et al. "PAYNT: A Tool for Inductive Synthesis of Probabilistic Programs". In: *Computer Aided Verification*. Ed. by Alexandra Silva and K. Rustan M. Leino. Cham: Springer International Publishing, 2021, pp. 856–869. ISBN: 978-3-030-81685-8. DOI: 10.1007/978-3-030-81685-8_40.

[4] Dana Angluin. "Learning Regular Sets from Queries and Counterexamples". In: *Inf. Comput.* 75.2 (1987), pp. 87–106.

[5] Thom S. Badings et al. "CTMCs with Imprecisely Timed Observations". In: *TACAS (2)*. Vol. 14571. Lecture Notes in Computer Science. Springer, 2024, pp. 258–278.

[6] Christel Baier and Joost-Pieter Katoen. *Principles of Model Checking*. MIT Press, 2008. ISBN: 978-0-262-02649-9.

[7] Johan Håstad. "Some optimal inapproximability results". In: *J. ACM* 48.4 (2001), pp. 798–859.

[8] Christian Hensel et al. "The Probabilistic Model Checker Storm". In: *Int J Softw Tools Technol Transfer* 24.4 (Aug. 1, 2022), pp. 589–610. ISSN: 1433-2787. DOI: 10.1007/s10009-021-00633-z. URL: https://doi.org/10.1007/s10009-021-00633-z (visited on 09/02/2024).

[9] Sebastian Junges, Hazem Torfah, and Sanjit A. Seshia. "Runtime Monitors for Markov Decision Processes". In: *Computer Aided Verification*. Ed. by Alexandra Silva and K. Rustan M. Leino. Cham: Springer International Publishing, 2021, pp. 553–576. ISBN: 978-3-030-81688-9. DOI: 10.1007/978-3-030-81688-9_26.

[10] Edi Muškardin et al. "AALpy: An Active Automata Learning Library". In: *Innovations Syst Softw Eng* 18.3 (Sept. 1, 2022), pp. 417–426. ISSN: 1614-5054. DOI: 10.1007/s11334-022-00449-3. URL: https://doi.org/10.1007/s11334-022-00449-3 (visited on 10/14/2024).

[11] Wouter Smeenk et al. "Applying Automata Learning to Embedded Control Software". In: *Formal Methods and Software Engineering*. Ed. by Michael Butler, Sylvain Conchon, and Fatiha Zaïdi. Cham: Springer International Publishing, 2015, pp. 67–83. ISBN: 978-3-319-25423-4. DOI: 10.1007/978-3-319-25423-4_5.

[12] Frits W. Vaandrager. "Model learning". In: *Commun. ACM* 60.2 (2017), pp. 86–95.

[13] Nan Yang et al. "Improving Model Inference in Industry by Combining Active and Passive Learning". In: *2019 IEEE 26th International Conference on*

Software Analysis, Evolution and Reengineering (SANER). 2019 IEEE 26th International Conference on Software Analysis, Evolution and Reengineering (SANER). Feb. 2019, pp. 253–263. DOI: 10.1109/SANER.2019.8668007. URL: https://ieeexplore.ieee.org/document/8668007 (visited on 01/23/2025).

A Description of benchmarks

B Results from ToVer Verification Experiments

C Results from ToVer Learning Experiments

Luko: TODO: write about plots

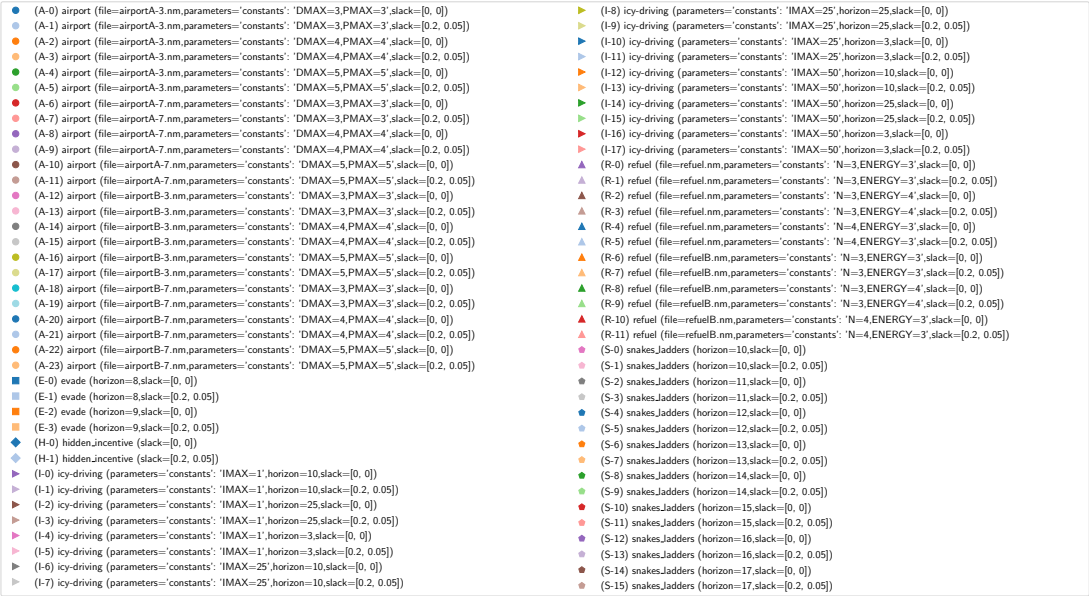


Fig. 14: Legend of symbols used in plots

		Benchmark						ToVer				Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
Airport	A-0	3/10	3/10	10	45	113	18	3983.53	128	0.30	0.30	172.76	183	0.30	0.30
Airport	A-1	1/10	7/20	10	45	113	18	3112.38	74	0.12	0.30	304.18	175	0.30	0.30
Airport	A-2	3/10	3/10	10	88	244	32	13530.51	222	0.30	0.30	311.28	238	0.30	1.00
Airport	A-3	1/10	7/20	10	88	244	32	9255.42	91	0.15	0.33	513.02	273	0.30	1.00

		Benchmark						ToVer				Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
Airport	A-4	3/10	3/10	10	145	423	50	5222.89	250	0.30	0.30	1029.72	326	0.30	1.00
Airport	A-5	1/10	7/20	10	145	423	50	3612.68	143	0.12	0.34	889.61	317	0.30	1.00
Airport	A-6	3/10	3/10	10	54	150	18	10992.00	154	0.30	0.30	309.27	228	0.30	0.30
Airport	A-7	1/10	7/20	10	54	150	18	2820.88	103	0.18	0.34	421.39	233	0.30	0.30
Airport	A-8	3/10	3/10	10	128	440	32	12964.04	381	0.30	0.30	2167.82	626	0.30	0.30
Airport	A-9	1/10	7/20	10	128	440	32	4275.92	128	0.15	0.33	1767.34	621	0.30	0.30
Airport	A-10	3/10	3/10	10	235	917	50	-	-	-	-	-	-	-	-
Airport	A-11	1/10	7/20	10	235	917	50	-	-	-	-	-	-	-	-
Airport	A-12	3/10	3/10	10	90	334	18	7838.16	129	0.30	0.30	293.90	168	0.30	0.30
Airport	A-13	1/10	7/20	10	90	334	18	4297.92	93	0.11	0.35	261.75	175	0.30	0.30
Airport	A-14	3/10	3/10	10	176	724	32	28299.34	217	0.30	0.30	1020.09	286	0.30	1.00
Airport	A-15	1/10	7/20	10	176	724	32	10621.05	106	0.11	0.32	703.50	255	0.30	1.00
Airport	A-16	3/10	3/10	10	290	1258	50	8544.17	242	0.30	0.30	1257.80	308	0.30	1.00
Airport	A-17	1/10	7/20	10	290	1258	50	4358.92	117	0.11	0.35	1348.91	313	0.30	1.00
Airport	A-18	3/10	3/10	10	108	432	18	19611.01	173	0.30	0.30	762.61	244	0.30	0.30
Airport	A-19	1/10	7/20	10	108	432	18	2602.35	98	0.16	0.33	852.17	243	0.30	0.30
Airport	A-20	3/10	3/10	10	256	1240	32	10403.48	416	0.30	0.30	3402.42	659	0.30	0.30
Airport	A-21	1/10	7/20	10	256	1240	32	11853.95	169	0.10	0.34	4108.96	673	0.30	0.30
Airport	A-22	3/10	3/10	10	470	2550	50	-	-	-	-	-	-	-	-
Airport	A-23	1/10	7/20	10	470	2550	50	-	-	-	-	-	-	-	-
Evade	E-0	3/10	3/10	8	385	1473	325	2214.51	218	0.30	0.24	2732.00	264	0.30	0.24
Evade	E-1	1/10	7/20	8	385	1473	325	944.43	144	0.30	0.31	2665.76	261	0.30	0.24
Evade	E-2	3/10	3/10	9	385	1473	325	4134.45	328	0.30	0.27	5320.22	369	0.30	0.27
Evade	E-3	1/10	7/20	9	385	1473	325	2534.33	181	0.30	0.35	5360.52	383	0.30	0.27
Hidden-Incentive	H-0	3/10	3/10	10	397	1649	100	9190.18	283	0.30	0.28	1448.18	348	0.30	0.43
Hidden-Incentive	H-1	1/10	7/20	10	397	1649	100	3740.06	243	0.17	0.34	1360.26	338	0.30	0.43
Icy-Driving	I-0	3/10	3/10	10	3	6	2	0.70	2	0.33	0.10	71.39	2	0.33	0.10
Icy-Driving	I-1	1/10	7/20	10	3	6	2	0.06	2	0.33	0.10	76.19	2	0.33	0.10
Icy-Driving	I-2	3/10	3/10	25	3	6	2	1750.09	2	0.33	0.10	175.51	2	0.33	0.10
Icy-Driving	I-3	1/10	7/20	25	3	6	2	0.07	2	0.33	0.10	172.98	2	0.33	0.10
Icy-Driving	I-4	3/10	3/10	3	3	6	2	0.14	8	0.33	0.10	27.81	2	0.33	0.10
Icy-Driving	I-5	1/10	7/20	3	3	6	2	0.03	2	0.33	0.10	28.28	2	0.33	0.10
Icy-Driving	I-6	3/10	3/10	10	27	125	2	102.05	5	0.42	0.30	92.19	5	0.42	0.30
Icy-Driving	I-7	1/10	7/20	10	27	125	2	61.50	5	0.42	0.30	91.48	5	0.42	0.30
Icy-Driving	I-8	3/10	3/10	25	27	125	2	-	-	-	-	-	-	-	-
Icy-Driving	I-9	1/10	7/20	25	27	125	2	-	-	-	-	-	-	-	-
Icy-Driving	I-10	3/10	3/10	3	27	125	2	0.20	6	1.00	0.19	28.60	5	1.00	0.19
Icy-Driving	I-11	1/10	7/20	3	27	125	2	0.04	5	1.00	0.19	30.03	5	1.00	0.19
Icy-Driving	I-12	3/10	3/10	10	52	250	2	102.18	5	0.39	0.28	90.23	5	0.39	0.28
Icy-Driving	I-13	1/10	7/20	10	52	250	2	95.72	5	0.39	0.28	93.22	5	0.39	0.28
Icy-Driving	I-14	3/10	3/10	25	52	250	2	-	-	-	-	-	-	-	-
Icy-Driving	I-15	1/10	7/20	25	52	250	2	-	-	-	-	-	-	-	-
Icy-Driving	I-16	3/10	3/10	3	52	250	2	0.14	6	1.00	0.19	30.86	5	1.00	0.19
Icy-Driving	I-17	1/10	7/20	3	52	250	2	0.04	5	1.00	0.19	31.43	5	1.00	0.19
Refuel	R-0	3/10	3/10	10	87	871	63	14.32	25	0.54	0.19	109.19	27	0.54	0.19
Refuel	R-1	1/10	7/20	10	87	871	63	15.68	27	0.54	0.19	110.84	27	0.54	0.19
Refuel	R-2	3/10	3/10	10	132	1798	72	30.38	32	0.32	0.14	121.21	30	0.32	0.14
Refuel	R-3	1/10	7/20	10	132	1798	72	201.44	31	0.32	0.14	120.02	31	0.32	0.14
Refuel	R-4	3/10	3/10	10	139	1628	107	52.75	40	0.54	0.19	139.53	41	0.54	0.19
Refuel	R-5	1/10	7/20	10	139	1628	107	52.02	42	0.54	0.19	140.19	42	0.54	0.19
Refuel	R-6	3/10	3/10	10	173	3429	63	24.68	27	0.54	0.19	121.45	25	0.54	0.19
Refuel	R-7	1/10	7/20	10	173	3429	63	26.42	27	0.54	0.19	123.68	26	0.54	0.19
Refuel	R-8	3/10	3/10	10	263	7127	72	54.80	29	0.32	0.14	139.59	30	0.32	0.14
Refuel	R-9	1/10	7/20	10	263	7127	72	368.99	31	0.32	0.14	143.13	30	0.32	0.14
Refuel	R-10	3/10	3/10	10	277	6415	107	97.30	43	0.54	0.19	162.26	40	0.54	0.19
Refuel	R-11	1/10	7/20	10	277	6415	107	120.52	42	0.54	0.19	188.77	43	0.54	0.19
SnL	S-0	3/10	3/10	10	101	502	4	2.44	24	0.85	0.21	304.14	33	0.85	0.21
SnL	S-1	1/10	7/20	10	101	502	4	1.92	23	0.85	0.21	317.51	31	0.85	0.21
SnL	S-2	3/10	3/10	11	101	502	4	6.43	42	0.40	0.21	404.83	64	0.40	0.21
SnL	S-3	1/10	7/20	11	101	502	4	11.39	42	0.40	0.21	390.89	62	0.40	0.21
SnL	S-4	3/10	3/10	12	101	502	4	97.99	78	0.39	0.29	1075.97	111	0.39	0.29

		Benchmark						ToVer				Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
SnL	S-5	1/10	7/20	12	101	502	4	68.09	64	0.29	0.29	812.45	114	0.39	0.29
SnL	S-6	3/10	3/10	13	101	502	4	291.77	129	0.31	0.29	448.13	178	0.31	0.29
SnL	S-7	1/10	7/20	13	101	502	4	155.05	150	0.31	0.29	1334.71	193	0.31	0.29
SnL	S-8	3/10	3/10	14	101	502	4	962.41	214	0.31	0.29	1259.73	295	0.31	0.29
SnL	S-9	1/10	7/20	14	101	502	4	1358.30	215	0.31	0.32	656.67	298	0.31	0.29
SnL	S-10	3/10	3/10	15	101	502	4	4995.18	343	0.31	0.29	1047.12	448	0.31	0.29
SnL	S-11	1/10	7/20	15	101	502	4	4797.06	310	0.31	0.32	1240.98	459	0.31	0.29
SnL	S-12	3/10	3/10	16	101	502	4	14325.03	501	0.30	0.29	860.93	638	0.30	1.00
SnL	S-13	1/10	7/20	16	101	502	4	15589.67	523	0.30	0.31	898.79	677	0.30	0.29
SnL	S-14	3/10	3/10	17	101	502	4	-	-	-	-	-	-	-	-
SnL	S-15	1/10	7/20	17	101	502	4	-	-	-	-	-	-	-	-

D Results from Verification Experiment

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert\mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert\mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
AirportA-3	A-0	3/10	10	MA	45	113	18	25	450	0.92	0.31	0.41	0.94
AirportA-3	A-1	-	10	MA	45	113	18	25	450	6.24	0.31	5.73	1.00
AirportA-3	A-2	3/10	10	FA	45	113	18	25	450	0.95	0.31	0.43	0.08
AirportA-3	A-3	-	10	FA	45	113	18	25	450	1014.53	0.31	1014.02	0.01
AirportA-3	A-4	3/10	10	MA	45	113	18	81	1458	85.24	0.35	84.64	1.00
AirportA-3	A-5	-	10	MA	45	113	18	81	1458	80.27	0.35	79.67	1.00
AirportA-3	A-6	3/10	10	FA	45	113	18	81	1458	1.41	0.36	0.81	0.14
AirportA-3	A-7	-	10	FA	45	113	18	81	1458	1310.78	0.35	1310.20	0.01
AirportA-3	A-8	3/10	10	MA	45	113	18	128	2304	833.83	0.56	832.93	-
AirportA-3	A-9	-	10	MA	45	113	18	128	2304	788.22	0.56	787.31	0.30
AirportA-3	A-10	3/10	10	FA	45	113	18	128	2304	1143.18	0.53	1142.35	-
AirportA-3	A-11	-	10	FA	45	113	18	128	2304	1106.29	0.55	1105.43	0.30
AirportA-3	A-12	3/10	10	MA	145	423	50	87	4350	3.71	1.58	1.38	1.00
AirportA-3	A-13	-	10	MA	145	423	50	87	4350	3.79	1.56	1.47	1.00
AirportA-3	A-14	3/10	10	FA	145	423	50	87	4350	11.48	1.56	9.20	0.30
AirportA-3	A-15	-	10	FA	145	423	50	87	4350	707.37	1.55	705.02	0.18
AirportA-3	A-16	3/10	10	MA	145	423	50	153	7650	4.84	2.23	1.74	1.00
AirportA-3	A-17	-	10	MA	145	423	50	153	7650	4.95	2.23	1.88	1.00
AirportA-3	A-18	3/10	10	FA	145	423	50	153	7650	14.34	2.22	11.28	0.30
AirportA-3	A-19	-	10	FA	145	423	50	153	7650	617.78	2.23	614.75	0.03
AirportA-3	A-20	3/10	10	MA	145	423	50	250	12500	219.17	3.52	214.56	-
AirportA-3	A-21	-	10	MA	145	423	50	250	12500	210.95	3.53	206.34	0.30
AirportA-3	A-22	3/10	10	FA	145	423	50	250	12500	487.30	3.51	482.82	-
AirportA-3	A-23	-	10	FA	145	423	50	250	12500	468.35	3.67	463.67	0.30
AirportA-7	A-24	3/10	10	MA	54	150	18	63	1134	18.60	0.53	17.74	0.32
AirportA-7	A-25	-	10	MA	54	150	18	63	1134	334.47	0.52	333.63	0.37
AirportA-7	A-26	3/10	10	FA	54	150	18	63	1134	2.79	0.50	2.01	0.29
AirportA-7	A-27	-	10	FA	54	150	18	63	1134	234.05	0.48	233.27	0.08
AirportA-7	A-28	3/10	10	MA	54	150	18	88	1584	107.41	0.78	106.15	0.33
AirportA-7	A-29	-	10	MA	54	150	18	88	1584	456.47	0.78	455.21	0.87
AirportA-7	A-30	3/10	10	FA	54	150	18	88	1584	4.07	0.81	2.80	0.17
AirportA-7	A-31	-	10	FA	54	150	18	88	1584	329.64	0.74	328.46	0.10
AirportA-7	A-32	3/10	10	MA	54	150	18	154	2772	1187.53	0.64	1186.37	-
AirportA-7	A-33	-	10	MA	54	150	18	154	2772	1103.16	0.66	1101.95	0.30
AirportA-7	A-34	3/10	10	FA	54	150	18	154	2772	443.97	0.64	442.87	-
AirportA-7	A-35	-	10	FA	54	150	18	154	2772	415.53	0.65	414.41	0.30
AirportA-7	A-36	3/10	10	MA	128	440	32	27	864	2.19	0.52	1.31	0.42
AirportA-7	A-37	-	10	MA	128	440	32	27	864	192.49	0.54	191.60	0.99
AirportA-7	A-38	3/10	10	FA	128	440	32	27	864	2.14	0.52	1.27	0.20
AirportA-7	A-39	-	10	FA	128	440	32	27	864	151.51	0.48	150.70	0.00
AirportA-7	A-40	3/10	10	MA	128	440	32	203	6496	12.55	1.14	10.63	0.68
AirportA-7	A-41	-	10	MA	128	440	32	203	6496	411.32	1.17	409.35	0.68

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
AirportA-7	A-42	3/10	10	FA	128	440	32	203	6496	4.84	1.15	2.96	0.13
AirportA-7	A-43	-	10	FA	128	440	32	203	6496	343.71	1.14	341.86	0.07
AirportA-7	A-44	3/10	10	MA	128	440	32	381	12192	624.65	2.73	620.57	-
AirportA-7	A-45	-	10	MA	128	440	32	381	12192	597.28	2.79	593.25	0.30
AirportA-7	A-46	3/10	10	FA	128	440	32	381	12192	1329.33	2.76	1325.47	-
AirportA-7	A-47	-	10	FA	128	440	32	381	12192	1306.39	2.72	1302.57	0.30
AirportB-3	A-48	3/10	10	MA	90	334	18	17	306	1.57	0.30	0.86	0.97
AirportB-3	A-49	-	10	MA	90	334	18	17	306	10.14	0.30	9.47	1.00
AirportB-3	A-50	3/10	10	FA	90	334	18	17	306	1.75	0.28	1.08	0.09
AirportB-3	A-51	-	10	FA	90	334	18	17	306	1501.45	0.30	1500.82	0.09
AirportB-3	A-52	3/10	10	MA	90	334	18	51	918	82.06	0.39	81.23	0.59
AirportB-3	A-53	-	10	MA	90	334	18	51	918	494.71	0.37	493.92	1.00
AirportB-3	A-54	3/10	10	FA	90	334	18	51	918	85.12	0.38	84.34	0.11
AirportB-3	A-55	-	10	FA	90	334	18	51	918	4822.51	0.38	4821.68	0.11
AirportB-3	A-56	3/10	10	MA	90	334	18	129	2322	1378.01	0.86	1376.39	-
AirportB-3	A-57	-	10	MA	90	334	18	129	2322	1361.23	0.86	1359.61	0.30
AirportB-3	A-58	3/10	10	FA	90	334	18	129	2322	3130.67	0.81	3129.17	-
AirportB-3	A-59	-	10	FA	90	334	18	129	2322	3016.85	0.92	3015.22	0.30
AirportB-3	A-60	3/10	10	MA	290	1258	50	68	3400	6.01	1.72	2.98	1.00
AirportB-3	A-61	-	10	MA	290	1258	50	68	3400	6.47	1.73	3.38	1.00
AirportB-3	A-62	3/10	10	FA	290	1258	50	68	3400	22.13	1.79	19.03	0.26
AirportB-3	A-63	-	10	FA	290	1258	50	68	3400	564.11	1.77	561.04	0.01
AirportB-3	A-64	3/10	10	MA	290	1258	50	168	8400	14.84	3.22	9.65	1.00
AirportB-3	A-65	-	10	MA	290	1258	50	168	8400	15.39	3.19	10.24	1.00
AirportB-3	A-66	3/10	10	FA	290	1258	50	168	8400	23.41	3.06	18.48	0.16
AirportB-3	A-67	-	10	FA	290	1258	50	168	8400	505.73	3.05	500.83	0.04
AirportB-3	A-68	3/10	10	MA	290	1258	50	242	12100	530.78	4.52	524.03	-
AirportB-3	A-69	-	10	MA	290	1258	50	242	12100	514.83	4.68	507.83	0.30
AirportB-3	A-70	3/10	10	FA	290	1258	50	242	12100	1266.89	4.60	1260.19	-
AirportB-3	A-71	-	10	FA	290	1258	50	242	12100	1335.06	4.71	1328.15	0.30
AirportB-7	A-72	3/10	10	MA	108	432	18	22	396	3.85	0.39	3.04	0.67
AirportB-7	A-73	-	10	MA	108	432	18	22	396	642.08	0.41	641.25	0.67
AirportB-7	A-74	3/10	10	FA	108	432	18	22	396	3.60	0.37	2.84	0.26
AirportB-7	A-75	-	10	FA	108	432	18	22	396	520.12	0.37	519.38	0.13
AirportB-7	A-76	3/10	10	MA	108	432	18	77	1386	568.81	0.57	567.46	0.32
AirportB-7	A-77	-	10	MA	108	432	18	77	1386	1343.02	0.57	1341.67	0.38
AirportB-7	A-78	3/10	10	FA	108	432	18	77	1386	4.64	0.56	3.34	0.24
AirportB-7	A-79	-	10	FA	108	432	18	77	1386	146.54	0.58	145.23	0.19
AirportB-7	A-80	3/10	10	MA	108	432	18	173	3114	2638.71	1.35	2635.97	-
AirportB-7	A-81	-	10	MA	108	432	18	173	3114	2542.38	1.32	2539.66	0.30
AirportB-7	A-82	3/10	10	FA	108	432	18	173	3114	478.38	1.28	475.86	-
AirportB-7	A-83	-	10	FA	108	432	18	173	3114	450.00	1.28	447.48	0.30
AirportB-7	A-84	3/10	10	MA	256	1240	32	20	640	3.71	0.72	2.34	0.69
AirportB-7	A-85	-	10	MA	256	1240	32	20	640	435.39	0.71	434.02	0.99
AirportB-7	A-86	3/10	10	FA	256	1240	32	20	640	57.80	0.71	56.47	0.23
AirportB-7	A-87	-	10	FA	256	1240	32	20	640	187.32	0.70	186.00	0.11
AirportB-7	A-88	3/10	10	MA	256	1240	32	249	7968	11.85	3.13	6.73	0.31
AirportB-7	A-89	-	10	MA	256	1240	32	249	7968	750.32	3.03	745.26	0.63
AirportB-7	A-90	3/10	10	FA	256	1240	32	249	7968	16.69	3.10	11.70	0.12
AirportB-7	A-91	-	10	FA	256	1240	32	249	7968	1341.91	3.24	1336.81	0.08
AirportB-7	A-92	3/10	10	MA	256	1240	32	416	13312	1142.12	5.55	1133.74	-
AirportB-7	A-93	-	10	MA	256	1240	32	416	13312	1124.68	5.30	1116.56	0.30
AirportB-7	A-94	3/10	10	FA	256	1240	32	416	13312	724.41	5.59	716.21	-
AirportB-7	A-95	-	10	FA	256	1240	32	416	13312	700.22	5.43	692.14	0.30
Evade	E-0	3/10	8	MA	385	1473	325	58	18850	29.20	19.91	4.35	1.00
Evade	E-1	-	8	MA	385	1473	325	58	18850	28.95	19.74	4.28	1.00
Evade	E-2	3/10	8	FA	385	1473	325	58	18850	28.46	20.17	4.37	-
Evade	E-3	-	8	FA	385	1473	325	58	18850	30.40	20.39	4.96	0.30
Evade	E-4	3/10	8	MA	385	1473	325	138	44850	43.84	34.65	3.97	1.00
Evade	E-5	-	8	MA	385	1473	325	138	44850	43.86	34.78	3.97	1.00
Evade	E-6	3/10	8	FA	385	1473	325	138	44850	42.39	34.89	3.97	-
Evade	E-7	-	8	FA	385	1473	325	138	44850	43.01	34.19	4.00	0.30
Evade	E-8	3/10	8	MA	385	1473	325	218	70850	48.76	42.16	3.41	-

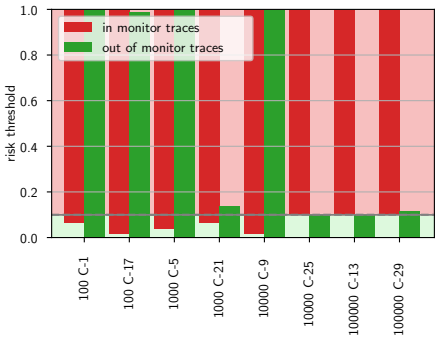
		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
Evade	E-9	-	8	MA	385	1473	325	218	70850	51.66	43.12	3.86	0.24
Evade	E-10	3/10	8	FA	385	1473	325	218	70850	49.04	42.65	3.37	-
Evade	E-11	-	8	FA	385	1473	325	218	70850	49.57	42.04	3.18	0.30
Evade	E-12	3/10	9	MA	385	1473	325	138	44850	38.70	20.02	9.13	1.00
Evade	E-13	-	9	MA	385	1473	325	138	44850	39.00	20.42	8.84	1.00
Evade	E-14	3/10	9	FA	385	1473	325	138	44850	37.63	20.30	9.05	-
Evade	E-15	-	9	FA	385	1473	325	138	44850	39.46	19.72	10.52	0.30
Evade	E-16	3/10	9	MA	385	1473	325	216	70200	49.36	31.65	8.41	1.00
Evade	E-17	-	9	MA	385	1473	325	216	70200	49.69	31.51	8.49	1.00
Evade	E-18	3/10	9	FA	385	1473	325	216	70200	48.44	32.21	8.46	-
Evade	E-19	-	9	FA	385	1473	325	216	70200	50.80	31.99	9.66	0.30
Evade	E-20	3/10	9	MA	385	1473	325	328	106600	49.11	33.35	8.23	-
Evade	E-21	-	9	MA	385	1473	325	328	106600	50.67	33.39	8.20	0.27
Evade	E-22	3/10	9	FA	385	1473	325	328	106600	49.75	34.26	8.14	-
Evade	E-23	-	9	FA	385	1473	325	328	106600	50.18	33.36	7.88	0.30
Hidden-Incentive	H-0	3/10	10	MA	397	1649	100	80	8000	14.67	5.20	5.52	1.00
Hidden-Incentive	H-1	-	10	MA	397	1649	100	80	8000	14.96	5.37	5.74	1.00
Hidden-Incentive	H-2	3/10	10	FA	397	1649	100	80	8000	16.20	5.63	6.80	0.14
Hidden-Incentive	H-3	-	10	FA	397	1649	100	80	8000	24.65	5.22	15.70	0.14
Hidden-Incentive	H-4	3/10	10	MA	397	1649	100	201	20100	17.67	8.47	5.67	1.00
Hidden-Incentive	H-5	-	10	MA	397	1649	100	201	20100	18.35	8.62	6.26	1.00
Hidden-Incentive	H-6	3/10	10	FA	397	1649	100	201	20100	21.21	8.31	9.60	0.22
Hidden-Incentive	H-7	-	10	FA	397	1649	100	201	20100	3123.98	8.40	3112.28	0.22
Hidden-Incentive	H-8	3/10	10	MA	397	1649	100	283	28300	1274.37	10.89	1260.42	-
Hidden-Incentive	H-9	-	10	MA	397	1649	100	283	28300	1290.20	10.90	1276.26	0.28
Hidden-Incentive	H-10	3/10	10	FA	397	1649	100	283	28300	6686.27	11.00	6672.29	-
Hidden-Incentive	H-11	-	10	FA	397	1649	100	283	28300	8822.49	10.86	8808.66	0.30
Icy-Driving	I-0	3/10	10	MA	3	6	2	2	4	0.02	0.01	0.01	-
Icy-Driving	I-1	-	10	MA	3	6	2	2	4	0.02	0.01	0.01	0.10
Icy-Driving	I-2	3/10	10	FA	3	6	2	2	4	0.55	0.01	0.54	-
Icy-Driving	I-3	-	10	FA	3	6	2	2	4	0.53	0.01	0.51	0.33
Icy-Driving	I-4	3/10	10	MA	3	6	2	2	4	0.02	0.01	0.01	-
Icy-Driving	I-5	-	10	MA	3	6	2	2	4	0.02	0.01	0.01	0.10
Icy-Driving	I-6	3/10	10	FA	3	6	2	2	4	0.56	0.01	0.55	-
Icy-Driving	I-7	-	10	FA	3	6	2	2	4	0.53	0.01	0.51	0.33
Icy-Driving	I-8	3/10	10	MA	3	6	2	2	4	0.02	0.01	0.01	-
Icy-Driving	I-9	-	10	MA	3	6	2	2	4	0.02	0.01	0.01	0.10
Icy-Driving	I-10	3/10	10	FA	3	6	2	2	4	0.55	0.01	0.54	-
Icy-Driving	I-11	-	10	FA	3	6	2	2	4	0.53	0.01	0.51	0.33
Icy-Driving	I-12	3/10	25	MA	3	6	2	2	4	0.03	0.01	0.01	-
Icy-Driving	I-13	-	25	MA	3	6	2	2	4	0.03	0.01	0.01	0.10
Icy-Driving	I-14	3/10	25	FA	3	6	2	2	4	1225.78	0.01	1225.76	-
Icy-Driving	I-15	-	25	FA	3	6	2	2	4	1173.95	0.01	1173.94	0.33
Icy-Driving	I-16	3/10	25	MA	3	6	2	2	4	0.03	0.01	0.01	-
Icy-Driving	I-17	-	25	MA	3	6	2	2	4	0.03	0.01	0.01	0.10
Icy-Driving	I-18	3/10	25	FA	3	6	2	2	4	1201.62	0.01	1201.60	-
Icy-Driving	I-19	-	25	FA	3	6	2	2	4	1215.28	0.01	1215.26	0.33
Icy-Driving	I-20	3/10	25	MA	3	6	2	2	4	0.03	0.01	0.01	-
Icy-Driving	I-21	-	25	MA	3	6	2	2	4	0.03	0.01	0.01	0.10
Icy-Driving	I-22	3/10	25	FA	3	6	2	2	4	1262.11	0.01	1262.10	-
Icy-Driving	I-23	-	25	FA	3	6	2	2	4	1204.32	0.01	1204.30	0.33
Icy-Driving	I-24	3/10	3	MA	3	6	2	8	16	0.01	0.00	0.01	-
Icy-Driving	I-25	-	3	MA	3	6	2	8	16	0.02	0.00	0.01	0.10
Icy-Driving	I-26	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-27	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33
Icy-Driving	I-28	3/10	3	MA	3	6	2	8	16	0.01	0.00	0.01	-
Icy-Driving	I-29	-	3	MA	3	6	2	8	16	0.02	0.00	0.01	0.10
Icy-Driving	I-30	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-31	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33
Icy-Driving	I-32	3/10	3	MA	3	6	2	8	16	0.01	0.00	0.01	-
Icy-Driving	I-33	-	3	MA	3	6	2	8	16	0.02	0.00	0.01	0.10
Icy-Driving	I-34	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-35	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
Icy-Driving	I-36	3/10	10	MA	27	125	2	5	10	0.75	0.02	0.73	-
Icy-Driving	I-37	-	10	MA	27	125	2	5	10	0.73	0.02	0.71	0.30
Icy-Driving	I-38	3/10	10	FA	27	125	2	5	10	99.46	0.02	99.44	-
Icy-Driving	I-39	-	10	FA	27	125	2	5	10	93.26	0.02	93.23	0.42
Icy-Driving	I-40	3/10	10	MA	27	125	2	5	10	0.73	0.02	0.71	-
Icy-Driving	I-41	-	10	MA	27	125	2	5	10	0.72	0.02	0.70	0.30
Icy-Driving	I-42	3/10	10	FA	27	125	2	5	10	98.13	0.02	98.11	-
Icy-Driving	I-43	-	10	FA	27	125	2	5	10	94.41	0.02	94.39	0.42
Icy-Driving	I-44	3/10	10	MA	27	125	2	5	10	0.73	0.02	0.71	-
Icy-Driving	I-45	-	10	MA	27	125	2	5	10	0.71	0.02	0.68	0.30
Icy-Driving	I-46	3/10	10	FA	27	125	2	5	10	97.12	0.02	97.10	-
Icy-Driving	I-47	-	10	FA	27	125	2	5	10	92.13	0.02	92.10	0.42
Icy-Driving	I-48	3/10	3	MA	27	125	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-49	-	3	MA	27	125	2	6	12	0.02	0.00	0.01	0.19
Icy-Driving	I-50	3/10	3	FA	27	125	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-51	-	3	FA	27	125	2	6	12	-	-	-	-
Icy-Driving	I-52	3/10	3	MA	27	125	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-53	-	3	MA	27	125	2	6	12	0.02	0.00	0.01	0.19
Icy-Driving	I-54	3/10	3	FA	27	125	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-55	-	3	FA	27	125	2	6	12	-	-	-	-
Icy-Driving	I-56	3/10	3	MA	27	125	2	6	12	0.02	0.00	0.02	-
Icy-Driving	I-57	-	3	MA	27	125	2	6	12	0.03	0.00	0.01	0.19
Icy-Driving	I-58	3/10	3	FA	27	125	2	6	12	0.01	0.00	0.00	-
Icy-Driving	I-59	-	3	FA	27	125	2	6	12	-	-	-	-
Icy-Driving	I-60	3/10	10	MA	52	250	2	5	10	0.81	0.02	0.79	-
Icy-Driving	I-61	-	10	MA	52	250	2	5	10	0.75	0.02	0.73	0.28
Icy-Driving	I-62	3/10	10	FA	52	250	2	5	10	102.11	0.02	102.09	-
Icy-Driving	I-63	-	10	FA	52	250	2	5	10	95.05	0.03	94.96	0.39
Icy-Driving	I-64	3/10	10	MA	52	250	2	5	10	0.72	0.02	0.70	-
Icy-Driving	I-65	-	10	MA	52	250	2	5	10	0.71	0.02	0.69	0.28
Icy-Driving	I-66	3/10	10	FA	52	250	2	5	10	102.30	0.02	102.27	-
Icy-Driving	I-67	-	10	FA	52	250	2	5	10	94.48	0.02	94.45	0.39
Icy-Driving	I-68	3/10	10	MA	52	250	2	5	10	0.75	0.02	0.72	-
Icy-Driving	I-69	-	10	MA	52	250	2	5	10	0.74	0.02	0.71	0.28
Icy-Driving	I-70	3/10	10	FA	52	250	2	5	10	102.25	0.02	102.23	-
Icy-Driving	I-71	-	10	FA	52	250	2	5	10	93.56	0.02	93.53	0.39
Icy-Driving	I-72	3/10	3	MA	52	250	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-73	-	3	MA	52	250	2	6	12	0.03	0.00	0.01	0.19
Icy-Driving	I-74	3/10	3	FA	52	250	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-75	-	3	FA	52	250	2	6	12	-	-	-	-
Icy-Driving	I-76	3/10	3	MA	52	250	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-77	-	3	MA	52	250	2	6	12	0.03	0.00	0.01	0.19
Icy-Driving	I-78	3/10	3	FA	52	250	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-79	-	3	FA	52	250	2	6	12	-	-	-	-
Icy-Driving	I-80	3/10	3	MA	52	250	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-81	-	3	MA	52	250	2	6	12	0.03	0.01	0.01	0.19
Icy-Driving	I-82	3/10	3	FA	52	250	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-83	-	3	FA	52	250	2	6	12	-	-	-	-
Refuel	R-0	3/10	10	MA	87	871	63	24	1512	4.00	1.22	1.72	1.00
Refuel	R-1	-	10	MA	87	871	63	24	1512	4.22	1.21	1.90	1.00
Refuel	R-2	3/10	10	FA	87	871	63	24	1512	3.80	1.20	1.49	-
Refuel	R-3	-	10	FA	87	871	63	24	1512	12.02	1.26	9.65	0.54
Refuel	R-4	3/10	10	MA	87	871	63	24	1512	3.98	1.18	1.72	1.00
Refuel	R-5	-	10	MA	87	871	63	24	1512	4.18	1.23	1.86	1.00
Refuel	R-6	3/10	10	FA	87	871	63	24	1512	3.73	1.23	1.44	-
Refuel	R-7	-	10	FA	87	871	63	24	1512	11.89	1.27	9.50	0.54
Refuel	R-8	3/10	10	MA	87	871	63	25	1575	3.62	1.24	1.34	-
Refuel	R-9	-	10	MA	87	871	63	25	1575	3.60	1.21	1.33	0.19
Refuel	R-10	3/10	10	FA	87	871	63	25	1575	3.91	1.31	1.50	-
Refuel	R-11	-	10	FA	87	871	63	25	1575	11.91	1.22	9.55	0.54
Refuel	R-12	3/10	10	MA	132	1798	72	31	2232	9.37	2.30	4.41	1.00
Refuel	R-13	-	10	MA	132	1798	72	31	2232	9.66	2.08	4.84	1.00
Refuel	R-14	3/10	10	FA	132	1798	72	31	2232	8.27	2.16	3.46	-
Refuel	R-15	-	10	FA	132	1798	72	31	2232	55.78	2.11	50.99	0.32

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
Refuel	R-16	3/10	10	MA	132	1798	72	31	2232	9.38	2.11	4.47	1.00
Refuel	R-17	-	10	MA	132	1798	72	31	2232	9.63	2.08	4.82	1.00
Refuel	R-18	3/10	10	FA	132	1798	72	31	2232	8.18	2.10	3.47	-
Refuel	R-19	-	10	FA	132	1798	72	31	2232	56.10	2.12	51.36	0.32
Refuel	R-20	3/10	10	MA	132	1798	72	32	2304	8.10	2.15	3.33	-
Refuel	R-21	-	10	MA	132	1798	72	32	2304	8.27	2.19	3.37	0.14
Refuel	R-22	3/10	10	FA	132	1798	72	32	2304	8.33	2.17	3.43	-
Refuel	R-23	-	10	FA	132	1798	72	32	2304	55.13	2.16	50.29	0.32
Refuel	R-24	3/10	10	MA	139	1628	107	39	4173	14.28	4.07	5.90	1.00
Refuel	R-25	-	10	MA	139	1628	107	39	4173	14.65	4.28	6.06	1.00
Refuel	R-26	3/10	10	FA	139	1628	107	39	4173	13.53	4.13	5.16	-
Refuel	R-27	-	10	FA	139	1628	107	39	4173	28.43	4.09	20.02	0.54
Refuel	R-28	3/10	10	MA	139	1628	107	39	4173	14.24	4.10	5.86	1.00
Refuel	R-29	-	10	MA	139	1628	107	39	4173	14.74	4.28	6.11	1.00
Refuel	R-30	3/10	10	FA	139	1628	107	39	4173	13.49	4.09	5.20	-
Refuel	R-31	-	10	FA	139	1628	107	39	4173	28.89	4.12	20.31	0.54
Refuel	R-32	3/10	10	MA	139	1628	107	40	4280	13.71	4.45	5.03	-
Refuel	R-33	-	10	MA	139	1628	107	40	4280	13.28	4.15	4.97	0.19
Refuel	R-34	3/10	10	FA	139	1628	107	40	4280	13.93	4.13	5.25	-
Refuel	R-35	-	10	FA	139	1628	107	40	4280	28.78	4.15	20.37	0.54
RefuelB	R-36	3/10	10	MA	173	3429	63	25	1575	7.48	1.83	3.50	1.00
RefuelB	R-37	-	10	MA	173	3429	63	25	1575	8.16	1.90	4.08	1.00
RefuelB	R-38	3/10	10	FA	173	3429	63	25	1575	7.06	1.88	3.00	-
RefuelB	R-39	-	10	FA	173	3429	63	25	1575	26.62	1.88	22.52	0.54
RefuelB	R-40	3/10	10	MA	173	3429	63	25	1575	7.52	1.85	3.50	1.00
RefuelB	R-41	-	10	MA	173	3429	63	25	1575	8.00	1.88	3.86	1.00
RefuelB	R-42	3/10	10	FA	173	3429	63	25	1575	7.09	1.83	3.04	-
RefuelB	R-43	-	10	FA	173	3429	63	25	1575	26.70	1.90	22.57	0.54
RefuelB	R-44	3/10	10	MA	173	3429	63	27	1701	6.79	1.83	2.79	-
RefuelB	R-45	-	10	MA	173	3429	63	27	1701	6.68	1.79	2.75	0.19
RefuelB	R-46	3/10	10	FA	173	3429	63	27	1701	6.96	1.78	3.02	-
RefuelB	R-47	-	10	FA	173	3429	63	27	1701	26.94	1.86	22.79	0.54
RefuelB	R-48	3/10	10	MA	263	7127	72	26	1872	17.77	3.88	8.72	1.00
RefuelB	R-49	-	10	MA	263	7127	72	26	1872	18.61	3.88	9.56	1.00
RefuelB	R-50	3/10	10	FA	263	7127	72	26	1872	15.95	3.97	6.81	-
RefuelB	R-51	-	10	FA	263	7127	72	26	1872	117.86	3.82	108.91	0.32
RefuelB	R-52	3/10	10	MA	263	7127	72	26	1872	18.11	4.08	8.74	1.00
RefuelB	R-53	-	10	MA	263	7127	72	26	1872	18.79	4.02	9.61	1.00
RefuelB	R-54	3/10	10	FA	263	7127	72	26	1872	15.83	3.82	6.70	-
RefuelB	R-55	-	10	FA	263	7127	72	26	1872	118.27	3.85	109.17	0.32
RefuelB	R-56	3/10	10	MA	263	7127	72	29	2088	16.08	4.19	6.74	-
RefuelB	R-57	-	10	MA	263	7127	72	29	2088	15.88	4.01	6.68	0.14
RefuelB	R-58	3/10	10	FA	263	7127	72	29	2088	16.11	3.99	6.87	-
RefuelB	R-59	-	10	FA	263	7127	72	29	2088	119.01	4.17	109.40	0.32
RefuelB	R-60	3/10	10	MA	277	6415	107	42	4494	26.23	5.43	12.13	1.00
RefuelB	R-61	-	10	MA	277	6415	107	42	4494	26.50	5.25	12.61	1.00
RefuelB	R-62	3/10	10	FA	277	6415	107	42	4494	24.31	5.45	10.34	-
RefuelB	R-63	-	10	FA	277	6415	107	42	4494	56.28	5.42	42.12	0.54
RefuelB	R-64	3/10	10	MA	277	6415	107	42	4494	25.77	5.39	11.96	1.00
RefuelB	R-65	-	10	MA	277	6415	107	42	4494	26.47	5.23	12.61	1.00
RefuelB	R-66	3/10	10	FA	277	6415	107	42	4494	24.42	5.17	10.65	-
RefuelB	R-67	-	10	FA	277	6415	107	42	4494	56.09	5.23	42.14	0.54
RefuelB	R-68	3/10	10	MA	277	6415	107	43	4601	24.03	5.42	10.07	-
RefuelB	R-69	-	10	MA	277	6415	107	43	4601	23.69	5.36	9.98	0.19
RefuelB	R-70	3/10	10	FA	277	6415	107	43	4601	24.83	5.49	10.60	-
RefuelB	R-71	-	10	FA	277	6415	107	43	4601	55.74	5.40	41.56	0.54
SnL	S-0	3/10	10	MA	101	502	4	24	96	0.63	0.11	0.48	-
SnL	S-1	-	10	MA	101	502	4	24	96	2.67	0.28	2.34	0.21
SnL	S-2	3/10	10	FA	101	502	4	24	96	0.55	0.11	0.41	-
SnL	S-3	-	10	FA	101	502	4	24	96	0.57	0.12	0.40	0.85
SnL	S-4	3/10	10	MA	101	502	4	24	96	0.72	0.20	0.48	-
SnL	S-5	-	10	MA	101	502	4	24	96	2.48	0.12	2.31	0.21
SnL	S-6	3/10	10	FA	101	502	4	24	96	0.57	0.12	0.41	-

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
SnL	S-7	-	10	FA	101	502	4	24	96	0.56	0.10	0.41	0.85
SnL	S-8	3/10	10	MA	101	502	4	24	96	0.64	0.11	0.49	-
SnL	S-9	-	10	MA	101	502	4	24	96	2.50	0.12	2.32	0.21
SnL	S-10	3/10	10	FA	101	502	4	24	96	0.55	0.10	0.42	-
SnL	S-11	-	10	FA	101	502	4	24	96	0.55	0.10	0.41	0.85
SnL	S-12	3/10	12	MA	101	502	4	1	4	-	-	-	-
SnL	S-13	-	12	MA	101	502	4	1	4	-	-	-	-
SnL	S-14	3/10	12	FA	101	502	4	1	4	-	-	-	-
SnL	S-15	-	12	FA	101	502	4	1	4	-	-	-	-
SnL	S-16	3/10	12	MA	101	502	4	46	184	17.99	0.25	17.64	-
SnL	S-17	-	12	MA	101	502	4	46	184	26.01	0.29	25.62	0.29
SnL	S-18	3/10	12	FA	101	502	4	46	184	1.75	0.26	1.41	0.00
SnL	S-19	-	12	FA	101	502	4	46	184	1.75	0.26	1.40	0.00
SnL	S-20	3/10	12	MA	101	502	4	78	312	22.58	0.44	22.02	-
SnL	S-21	-	12	MA	101	502	4	78	312	31.09	0.45	30.50	0.29
SnL	S-22	3/10	12	FA	101	502	4	78	312	3.73	0.42	3.21	-
SnL	S-23	-	12	FA	101	502	4	78	312	3.71	0.43	3.14	0.39
SnL	S-24	3/10	14	MA	101	502	4	1	4	-	-	-	-
SnL	S-25	-	14	MA	101	502	4	1	4	-	-	-	-
SnL	S-26	3/10	14	FA	101	502	4	1	4	-	-	-	-
SnL	S-27	-	14	FA	101	502	4	1	4	-	-	-	-
SnL	S-28	3/10	14	MA	101	502	4	139	556	12.32	1.04	10.88	0.82
SnL	S-29	-	14	MA	101	502	4	139	556	175.94	1.02	174.58	1.00
SnL	S-30	3/10	14	FA	101	502	4	139	556	37.99	0.96	36.70	0.29
SnL	S-31	-	14	FA	101	502	4	139	556	112.18	1.01	110.84	0.14
SnL	S-32	3/10	14	MA	101	502	4	214	856	152.92	1.69	150.95	-
SnL	S-33	-	14	MA	101	502	4	214	856	150.38	1.52	148.57	0.29
SnL	S-34	3/10	14	FA	101	502	4	214	856	183.23	1.47	181.55	-
SnL	S-35	-	14	FA	101	502	4	214	856	246.79	1.55	244.99	0.31
SnL	S-36	3/10	16	MA	101	502	4	1	4	-	-	-	-
SnL	S-37	-	16	MA	101	502	4	1	4	-	-	-	-
SnL	S-38	3/10	16	FA	101	502	4	1	4	-	-	-	-
SnL	S-39	-	16	FA	101	502	4	1	4	-	-	-	-
SnL	S-40	3/10	16	MA	101	502	4	269	1076	24.87	1.99	22.26	0.31
SnL	S-41	-	16	MA	101	502	4	269	1076	384.38	2.14	381.55	1.00
SnL	S-42	3/10	16	FA	101	502	4	269	1076	9.05	1.85	6.68	0.00
SnL	S-43	-	16	FA	101	502	4	269	1076	9.22	1.88	6.82	0.00
SnL	S-44	3/10	16	MA	101	502	4	501	2004	1357.81	7.91	1349.03	-
SnL	S-45	-	16	MA	101	502	4	501	2004	1235.13	8.16	1226.04	0.29
SnL	S-46	3/10	16	FA	101	502	4	501	2004	2723.58	7.94	2714.98	-
SnL	S-47	-	16	FA	101	502	4	501	2004	2536.77	7.72	2528.44	0.30

E Results from Baseline Sampling Count Experiment



(a) Risk threshold of 0.1

(b) Risk threshold of 0.2

(c) Risk threshold of 0.4

Fig. 15: Found risk thresholds when using baseline leaning with a different amount of samples in each EQ step. This experiment is done for three different risk thresholds.